Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

Contents

*a.gori23@studenti.unipi.it / nepero27178@github.com

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_{c}	Critical temperature

Chapter 1

The normal phase

This chapter is devoted to the MFT analysis of the “normal phase”, which is, the one where interacting electrons do not develop an antiferromagnetic or superconducting gap, although undergoing an effective non-trivial transformation. What will be derived here, in particular the hopping renormalization effect, constitutes a peculiarity of effective MFT treatment of the EHM and will be the groundwork of following analyses. Note that this chapter is structured in such a way that what is derived here mathematically is recovered and expanded to more complex phases in next chapters.

1.1 The “normal” phase and its symmetries

The normal phase is simply the one where no symmetry is broken, and the many-body ground state preserves the entire U(2) structure of Eq. (**??**) as well as full translational symmetry $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. In terms of spatial symmetries, we here theoretically allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

symmetry reduction and extract the mathematical interesting consequences. While this implies immediately inversion symmetry preservation, full C_4 symmetry will be recovered later. Many materials show interactions-driven nematicity, which is the capability of partially breaking rotational symmetry.

The normal phase is essentially given by the null gap limit of both the AF and SC phases. By discussing it separately now, we highlight a MFT feature of the EHM of particular interest, namely, the insurgence of a new, self-consistent hopping amplitude. From Tabs. **??** and **??**, we know that the relevant Wick's contractions (RWCs) for this phase are

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the final effects of all RWCs. As we will see in detail and is reported in Tab. 1.1, the Hartree contractions account essentially for chemical potential shifts, while the only Fock contraction for a renormalization effects on the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

1.2 Hartree effects: chemical potential renormalization

The first RMP we discuss is μ, whose numerical value is renormalized by Hartree channel interactions within MFT. As is discussed in Sec. **??**, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently; thus any net effect that shifts μ

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 1.1 | Relevant Wick's contractions and net effect in the normal phase.

is effectively ignored within (and limitedly to) the iterative algorithm. Renormalization effects become relevant when computing free energy.

1.2.1 Hartree channel in the normal phase: analytic derivation

Within this section we discuss analytically the effects of MFT on the Hartree channel, separating the local and non-local interactions and obtaining a set of HFPs needed to setup an interative HF algorithm.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{1.1}$$

which gives immediately

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term) due to the Hartree (H) contraction of the U repulsion, discussed in Sec. 1.2.2. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{1.2}$$

The non-local attraction further corrects μ.

Non-local attraction. Let us break down $\hat{H}_V$ isolating its Hartree RWCs in both o.s. and s.s. sectors:

$$\begin{aligned}\hat{H}_V \simeq &\overbrace{-V \sum_{\langle ij\rangle} \sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma}\langle \hat{n}_{j\sigma}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ &\underbrace{-V \sum_{\langle ij\rangle} \sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\langle \hat{n}_{j\overline{\sigma}}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right]}_{\text{o.s.}} + (\text{all the rest})\end{aligned} \tag{1.3}$$

where "all the rest" collects all non-Hartree RWCs. Recalling the Ansatz of Eq. (1.1), we get

$$\hat{H}_V \simeq \underbrace{-nV \sum_{\langle ij\rangle} \sum_\sigma \left[\hat{n}_{j\sigma} + \hat{n}_{i\sigma}\right]}_{\text{s.s.}} \underbrace{-nV \sum_{\langle ij\rangle} \sum_\sigma \left[\hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\right]}_{\text{o.s.}} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Now, evidently both sums above are just a number operator, thus including the degeneracies introduced by the summations on NNs for a square lattice with coordination number $z = 4$

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{1.4}$$

Considering the net shift to μ due to both interactions by combining Eqns. (1.2), (1.4), we get the final RMP anticipated in Tab. 1.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.5}$$

This result remains valid throughout all phases discussed in this text: for the AF phase, the SDW character leaves this shift untouched, while the SC phase we will discuss is inherently translational invariant.

1.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle \\ &= L_x L_y \times n^2 U \end{aligned} \tag{1.6}$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle + \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\bar{\sigma}}\rangle\right] \\ &= -2V \sum_{\langle ij\rangle}\sum_{\sigma} n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \tag{1.7}$$

for the non-local attraction. In the last passage, we used Eq. (**??**), as well as spin degeneracy. Note that, correctly, the contraction energy is an extensive quantity. This implies that we can define a contraction energy per site – something we will do later when computing total free energy density.

1.3 Fock effects: bare bands renormalization

Up to now, no relevant changes have been made to the starting situation of the conventional degenerate Fermi gas. The scenario changes now, with the analysis of the Fock RWCs. Even if no gap opens, in the normal phase particles interact by the means of the non-local attraction $\hat{H}_V$, a feature that effectively redefines the bare fermionic bands.

From Wick's decomposition of $\hat{H}_V$, the only allowed Fock term comes from the same-spin part due to SU(2) symmetry selection rules. Let us focus only on this term when decomposing $\hat{H}$:

$$\hat{H}_V \simeq (\text{Hartree channel}) + V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right] \tag{1.8}$$

(note the + sign in front of the displayed term). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \text{h.c.}\right] \\ &= -\sum_{\langle ij\rangle}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

1.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (**??**), applied to the displayed part of Eq. (1.8),

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.9}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.10}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands, we get

$$\langle\hat{c}^{\dagger}_{\mathbf{k}_1\sigma_1}\hat{c}_{\mathbf{k}_2\sigma_2}\rangle=\delta_{\mathbf{k}_1=\mathbf{k}_2}\delta_{\sigma_1=\sigma_2}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \tag{1.11}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon;\beta,\mu)\equiv\frac{1}{e^{\beta(\epsilon-\mu)}+1}$$

In other words, we are assuming a solution to exist, we call "normal", such that the net effect of the interactions is to generate an effective correction to bands such that the final hamiltonian is just $\hat{H}\simeq\hat{H}_{\tilde{t}}+$(shifts). It follows that Eq. (1.9) becomes

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.12}$$

since the only contribution comes from $\mathbf{k}'=-\mathbf{k}$.

Let now $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$, $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. A representation of this variables change is given in Fig. 1.1. Being for $\ell=x,y$

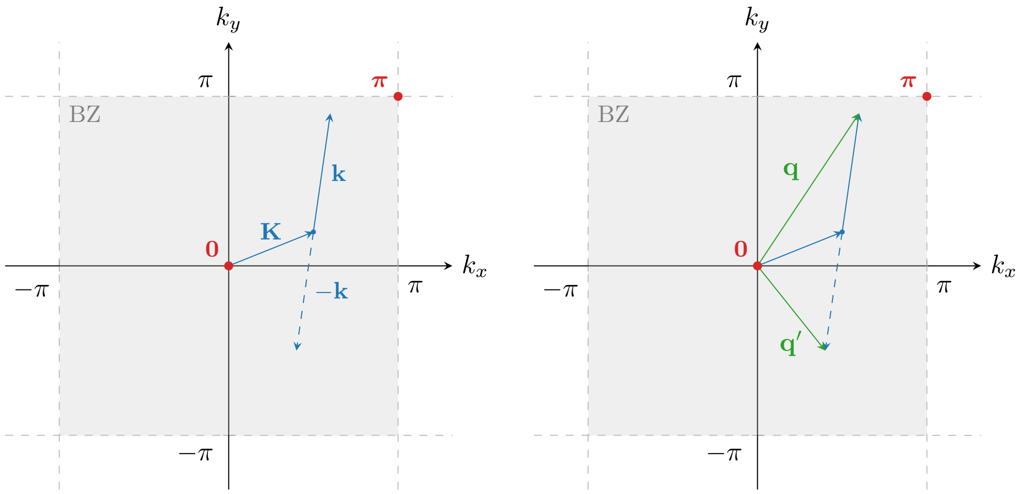
$$\begin{aligned}\delta q_\ell&\equiv q_\ell-q'_\ell\\&=(K_\ell+k_\ell)-(K_\ell-k_\ell)\\&=2k_\ell\end{aligned}$$

we reduce Eq. (1.9) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos(\delta q_x)+\cos(\delta q_y)\right]\langle\hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.13}$$

Recall now the result of the symmetry channels decomposition of Eq. (**??**),

$$\cos(\delta q_x)+\cos(\delta q_y)=\frac{1}{2}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\quad\text{for }\gamma\in\{s^*,p_x,p_y,d_{x^2-y^2}\}$$



a | Generic vectors. b | Change of variables.

Figure 1.1 | Representation of the variable change employed to pass from Eq. (1.12) to (1.13). In Fig. 1.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 1.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Thanks to this handy property, we reduce Eq. (1.13) to

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\stackrel{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \qquad (1.14)$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \qquad (1.15)$$

Of course, the numerical value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution; however, for the sake of reproducibility of this derivation in next chapters, we omit that for a moment. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Now, the bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric, but in general $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. We assume respected spatial inversion invariance, so it must be

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

thus

$$u^{(p_\ell)}_\sigma = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

recalling the antisymmetric form factors of Tab **??**. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 1.2 using spin degeneracy. We anticipate here that, as will result from Sec. 1.5.2, that the d-wave component will be suppressed. To prove this statement requires a little space, thus we will deal with it later on. For now we keep this component ignoring these anticipations, for reasons that will become clear in Sec. 1.3.3.

The two HFPs $u^{(s^*)}_\sigma$ and $u^{(d)}_\sigma$ are in general non-zero, since we are allowing theoretically the possibility for $\mathrm{C}_4 \to \mathrm{C}_2$ SSB. Note that when imposing C_4 invariance the $u^{(d)}_\sigma$ HFP vanishes, while

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??

Table 1.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

in general $u_\sigma^{(s^*)}$ is non-zero. From these considerations Eq. (1.14) becomes

$$V \sum_{\langle ij\rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\stackrel{\mathcal{F}}{=} V \sum_{\mathbf{q}\in\mathrm{BZ}} \sum_{\sigma} \left[u_\sigma^{(s^*)} \varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)} \varphi_{\mathbf{q}}^{(d)*} \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.16}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V (\cos k_x + \cos k_y) + u_\sigma^{(d)} V (\cos k_x - \cos k_y) \tag{1.17}$$

This equation defines the bands renormalization and resymmetrization anticipated in Tab. 1.1, and is actually spin-independent. The fact that both a s^* and a d-wave component appear introduces an interesting anisotropy that is discussed in next section.

1.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (1.10):

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{F}/V} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u^*_{\mathbf{q}'\sigma} \right] \\ &= L_x L_y \times \frac{V}{2} \sum_{\sigma} \left[\left|u_\sigma^{(s^*)}\right|^2 + \left|u_\sigma^{(d)}\right|^2 \right] \end{aligned} \tag{1.18}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The particularly relevant result, here, is the fact that the full spectral amplitude of $u_{\mathbf{k}\sigma}$ here is weighted and contributes to contraction energy with a V linear prefactor. This suggests that a mixed-symmetry band could be energetically preferred.

1.3.3 Nematicity: extending hopping to d-wave bands

This section serves as a zoom on the subject of free band structures topologies, and to justify the interest in a $C_4 \to C_2$ SSB apparently pretty arbitrary. [Insert: nematicity is a common feature in many materials, so on and so forth...] From Tabs. ?? and ?? we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma\in\{s^*,d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

Interestingly enough, in principle an anisotropy in the normal diffusion between x and y direction emerges: such a feature is referred to as “nematicity”. A schematics of anisotropic mixed symmetry

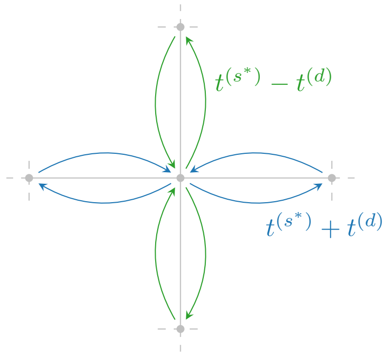


Figure 1.2 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

hopping is in Fig. 1.2: as is evident, for each bond linking two sites the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. Naturally, the solution needs to be invariant under exchange $x \leftrightarrow y$, making the actual SSB of the class $C_4 \to C_2/\mathbb{Z}_2$. For the normal phase, the physical expectation is that nematicity is suppressed; this does not make this discussion somewhat less relevant for more complex phases, such as the superconducting one, for which it is known nematicity in 2D planar lattices can arise as a source of superconducting long-range ordering [3].

The fact that hopping is here anisotropic does not break full translational invariance $T_1^{(x)} \otimes T_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (1.11) necessarily gives a uniform density indeed, whatever the shape of the band. By Fourier-transforming the site number operator according to the convention of Eq. (**??**),

$$\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y}\sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}$$

which obviously implies

$$\langle \hat{n}_{i\sigma}\rangle = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that: C_4 SSB is not manifested this way.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, in vector notation and recalling Tab. **??**:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$$

The fact that $t/2$ appears here instead of t is not inconsistent, it's just a consequence of the equivalence:

$$-t\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right] = -\frac{t}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right]$$

The diffusion RMP $\tilde{t}_{ij\sigma}$ is to be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2} V \tag{1.19}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2} V \tag{1.20}$$

Both components are controlled by V. Evidently $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the hopping anisotropy,

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V} \qquad u^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{V}$$

Theoretical introduction to nematicity is over, and we will get back to this later. To complete the phase overview, we move to the computation of the ground-state free energy density.

1.4 Normal free energy density

An important quantity to derive in order to compare different phases is free energy density. The mere existence of a MFT solution to a given phase is not enough to understand if the latter is physically established. To compare energy is instead the correct method to understand what phase is dominant within a given set. In this section we sketch a procedure to extract free energy density we will use identically everywhere. Let

$$f^{(\mathrm{N})}_{\mathrm{MFT}} \equiv \frac{F^{(\mathrm{N})}_{\mathrm{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k},\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} + E^{(\mathrm{N})}_{\mathrm{c}} - \tilde{\mu} \hat{N}$$

where $E^{(\mathrm{N})}_{\mathrm{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\mathrm{N})}_{\mathrm{c}} \equiv - \left[E^{(\mathrm{N})}_{\mathrm{H}/U} + E^{(\mathrm{N})}_{\mathrm{H}/V} + E^{(\mathrm{N})}_{\mathrm{F}/V} \right]$$

having included the various contraction energies from Eqns. (1.6), (1.7) and (1.18).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 1.1. For a free Fermi gas, the free energy density is simply obtained by summing four contributions:

1. The free particles thermal ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e^{(\mathrm{N})}_{\mathrm{c}} &\equiv \frac{E^{(\mathrm{N})}_{\mathrm{c}}}{L_x L_y} \\ &= -n^2 U + V \left[2zn^2 - \left| u^{(s^*)} \right|^2 - \left| u^{(d)} \right|^2 \right] \end{aligned}$$

3. The entropic contribution to free energy

$$e^{(\mathrm{N})}_{s} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [8],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{1.21}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$
$$- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big]$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. ??.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + [\mu + n(2zV - U)]\, n \end{aligned}$$

When it comes to algorithmic computation of this quantity, it is important to express μ in terms of its RMP counterpart $\tilde{\mu}$, as discussed in Sec. ??. By doing this we are considering all the effects of MFT decomposition of $\hat{H}$. However, as was predicted, the RMP $\tilde{\mu} = \mu + n(2zV - U)$ presented in Eq. (1.5) appears because of contraction energy corrections, giving finally

$$f_{\mathrm{MFT}}^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu} n \tag{1.22}$$

a simple enough formula including all effects. Note that, in computational therms, logarithmic divergences at null argument (occurring in the formula above at energies deep in the Fermi sea) can be difficult to handle. It is then useful to implement some algebraic manipulation to overcome this difficulty in numeric computations.

1.5 Theoretical constraints on the normal phase HFPs

Within this section we discuss some theoretical constraints on the two HFPs identified in this section. These last are useful in later numerical considerations. The discussion is hereby divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and SSB.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let in general for notational simplicity

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

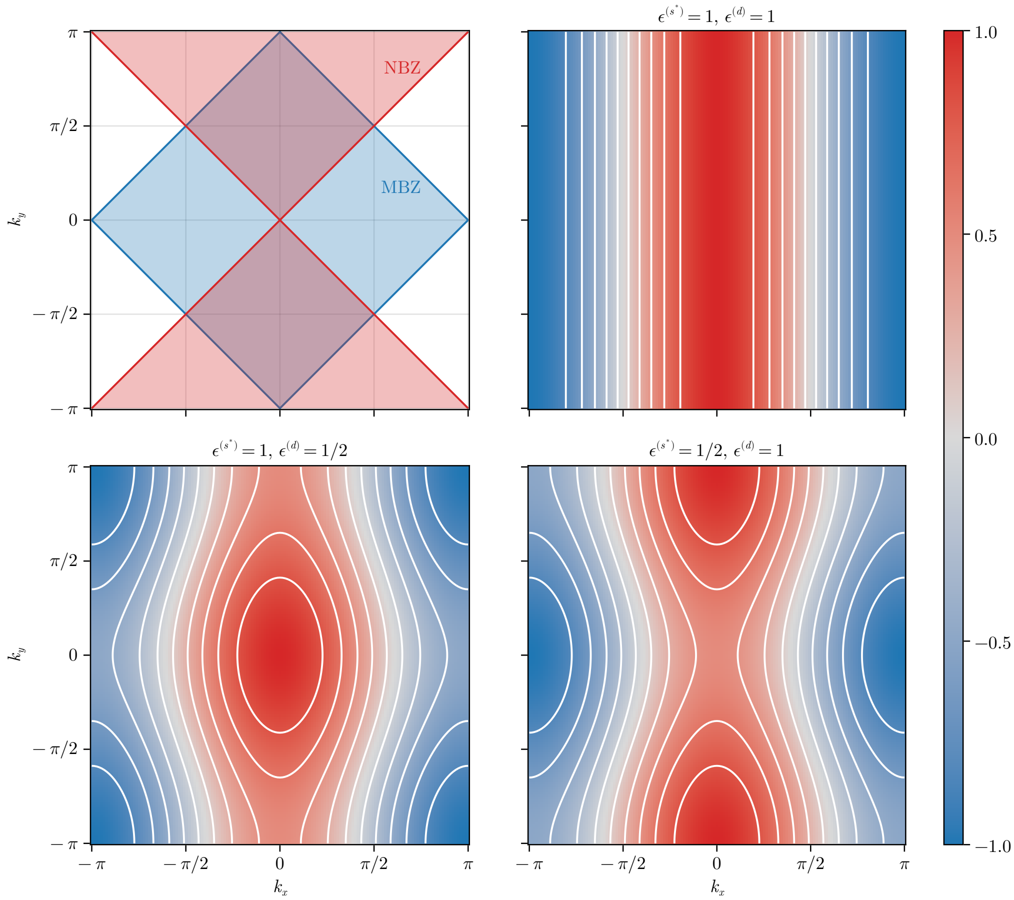


Figure 1.3 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalized band. Consider the s^*-wave and d-wave structure factors already presented in Fig. **??**, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\}$$
$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\}$$

represented in the top-left panel of Fig. 1.3. In the other panels are reported examples of mixed symmetry bands, for positive components. The respective opposite sign counterparts are obtained easily by the following rules:

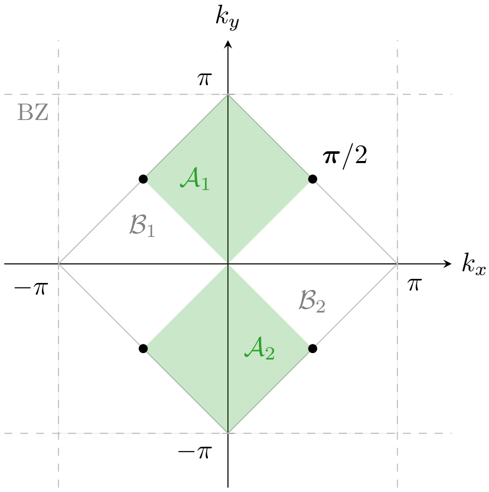
$$\begin{aligned}
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Invert colors in Fig. 1.3} \\
&\epsilon^{(s^*)} > 0 \quad \text{and} \quad \epsilon^{(d)} < 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ in Fig. 1.3} \\
&\epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} > 0 \quad \Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 1.3}
\end{aligned}$$

It is useful to concentrate on the MBZ and the NBZ, being various computations later performed precisely over these domains. Starting from the intersections of the two magnetic zones, both can be factored in two sectors of interest,

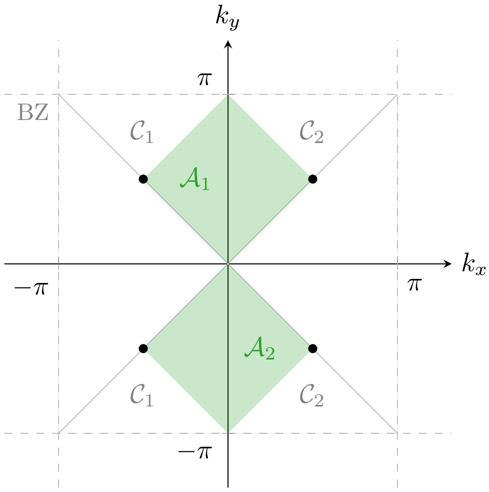
$$\text{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \text{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 1.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$



a | MBZ sectors.



b | NBZ sectors.

Figure 1.4 | Depiction of the sectors composing the MBZ (Fig. 1.4a) and the NBZ (Fig. 1.4b), based on the divisions shown in the top-left panel of Fig. 1.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

These domains satisfy:

$$
\begin{aligned}
\mathcal{A} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\
&= \mathrm{MBZ} \cap \mathrm{NBZ} \\
\mathcal{B} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \\
&= \mathrm{MBZ} \setminus \mathrm{NBZ} \\
\mathcal{C} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \quad \text{and} \quad \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\
&= \mathrm{NBZ} \setminus \mathrm{MBZ}
\end{aligned}
$$

and will be used later on. Importantly, whatever the shape of the mixed band, the following equation always holds

$$
|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \Longrightarrow \quad \tilde{\epsilon}_{(k_x,k_y)} = 0
$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 1.4.

1.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalized bands are

$$
\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)} V - 2t)(\cos k_x + \cos k_y)
$$

with a single self-consistency equation

$$
u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
$$

Just by looking at these two equations we can conclude:

1. For $t > 0$, there is no stable solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. A generic sum over the BZ is identically written as

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}\right]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu})\right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right] \end{aligned} \tag{1.23}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalized band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no stable solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalized bands is positive. This implies the renormalized band $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

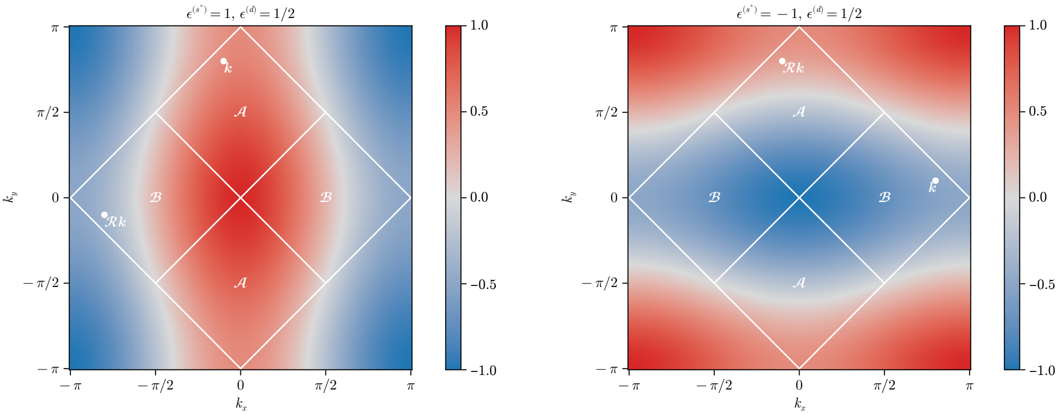
which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{1.24}$$

Interestingly enough, the self-consistent procedure predicts a positive convergence value for the simplest HFP, excluding the possibility of nullifying the effective hopping as well as bands sign inversion. We can denominate this net effect “bands shrinking”.



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 1.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eq. (1.25).

1.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (1.17). The proofs of this section are somewhat more involuted, and lead to the conclusion that within the normal phase the C_4 structure must be preserved, leading us back to the derivation of the previous section.

1. It is easy to show that a pure d-wave band is impossible to realize at any finite $t > 0$: if the band is purey d-wave, then the relative Fermi-Dirac distribution cannot admit s^*-wave components,

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) = 0$$

   thus it must be $u^{(s^*)} = 0$. We have

$$u^{(s^*)} = 2t/V \quad \Longrightarrow \quad u^{(s^*)} = 0$$

   which is absurd at finite values. A purely d-wave band is impossible.

2. Let for instance

$$u^{(s^*)} > 2t/V > 0 \qquad u^{(d)} > 0$$

   The resulting band is analogous to the ones of Fig. 1.3. Within this structure and using the notation of Fig. 1.4a, it always holds

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

   Recollecting Eq. (1.23) which is identically written as

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$
$$+ \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

   The sum with $\mathcal{B}$ requires a little care: in fact, as is evident from the bottom-left panel of

Fig. 1.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{1.25}$$

A graphical rendition of this idea is given in Fig. 1.5a.

Now: both $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ are positive by construction, then it holds

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad |\tilde{\epsilon}_{\mathbf{k}}| \geq |\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}|$$

which a trivial consequence of the fact that $|a-b| \leq |a+b|$ for positive a, b and

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\geq 0} \end{aligned}$$

Now, the Fermi-Dirac distribution is monotonically descendent: this means that

$$\begin{aligned} -\tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \quad &\Longrightarrow \quad f(-\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}};\beta,\tilde{\mu}) \\ -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq \tilde{\epsilon}_{\mathbf{k}} \quad &\Longrightarrow \quad f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}};\beta,\tilde{\mu}) \geq f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \end{aligned}$$

thus we have

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} &(\cos k_x + \cos k_y) \\ &\times \left[f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}};\beta,\tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}};\beta,\tilde{\mu})\right] \end{aligned}$$

which leads to an absurd,

$$u^{(s^*)} > 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

3. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to exclude the possibility of
$$u^{(s^*)} > 2t/V > 0 \qquad u^{(d)} < 0$$
thus we can self-consistently claim $u^{(s^*)} < 2t/V$.

4. Let now
$$u^{(s^*)} < 0 \qquad u^{(d)} > 0$$
Within this structure, it always holds
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$
Proceed exactly as before, with identical meaning of the notation and inverted roles for $\mathcal{A}$ and $\mathcal{B}$, as drawn in Fig. 1.5b. First, we have
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad |\tilde{\epsilon}_{\mathbf{k}}| \leq |\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}|$$
which a trivial consequence of the fact that $|a-b| \geq |a+b|$ for $a \leq 0$ and $b \geq 0$, and
$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\leq 0} \end{aligned}$$
By repeating identical steps as before we end up with the implication
$$u^{(s^*)} < 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$
which is absurd.

5. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to exclude the possibility of
$$u^{(s^*)} < 0 \qquad u^{(d)} < 0$$

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (1.24) needs to be fulfilled. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

6. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads
$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$
where we recognized, as is shown in Fig. 1.4b, that $\mathrm{BZ} \setminus \mathrm{NBZ} = \mathcal{R}\mathrm{NBZ}$. It holds:
$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$
thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get
$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$
which, as always, is absurd.

7. An identical argument, with inverted inequalities, can be carried out to exclude the possibility of
$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

These last two facts were the missing informations. Importantly, self-consistency excludes the possibility of a $C_4 \to C_2$ SSB in the normal phase, giving the constraint
$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{1.26}$$
A consequence of this is that there is no need at all to bother using $u^{(d)}$ as an HFP. This is physically expected but non trivial.

At this point, bands shrinking is a guaranteed effect, and the point to be asked is where the converged value of $u^{(s^*)}$ will be positioned in its admissibility range, based on the model parameter. This question is the subject of next sections.

1.5.3 Expected dependence on δ at low temperature

By looking once again at the self-consistency equation, we can infer roughly the physical expectation for $u^{(s^*)}$ when the system is subject to electronic doping. Consider first the low-temperature limit, namely $\beta \gg 1$. Consider the self-consistency equation of Tab. 1.2 seen as a functional of temperature and chemical potential,
$$u^{(s^*)}[\beta, \tilde{\mu}] = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$
in the zero-temperature limit, as long as the shrunk band is not flat, it holds
$$\lim_{\beta\to\infty} u^{(s^*)}[\beta, \tilde{\mu}] = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Now, at low temperature the Fermi-Dirac slope becomes less steep, a fact that makes the above equation only approximately verified as long as the convergence value of $u^{(s^*)}$ results in a "not-so-strong" bands shrinking.

Let us assume the latter, which means we assume it holds

$$u^{(s^*)}[\beta,\tilde{\mu}]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\epsilon_{\mathbf{k}}\leq\tilde{\mu}}(\cos k_x + \cos k_y)$$

which, at this point, will be true only in an unspecified region of parameters space. Of course, the "not-so-strong" shrinking assumption implies the fact that a FS is well-defined, being the band non-flat. At half filling, because of PHS,

$$\delta = 0 \qquad \Longrightarrow \qquad \mathrm{FS} = \partial\mathrm{MBZ}$$

With $\partial\Gamma$ we indicate the contour of the simply connected domain Γ. At $\delta > 0$ the FS expands over the MBZ boundary, where the form factor $\cos k_x + \cos k_y$ becomes negative. This gives the self-consistency equation

$$u^{(s^*)}[\beta,\tilde{\mu}]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\epsilon_{\mathbf{k}}\leq 0}|\cos k_x + \cos k_y| - \frac{1}{L_x L_y}\sum_{0<\epsilon_{\mathbf{k}}\leq\tilde{\mu}}|\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximized whenever $\tilde{\mu} = 0$, at half filling.

This holds as long as the effective hopping is non-zero, a condition implying together

$$\delta_1 > \delta_2 \qquad \Longrightarrow \qquad \mathrm{FS}_1 \text{ encircles } \mathrm{FS}_2$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \qquad \Longrightarrow \qquad \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde{\mu}]\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde{\mu}]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is "not-so-strong" to rise the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

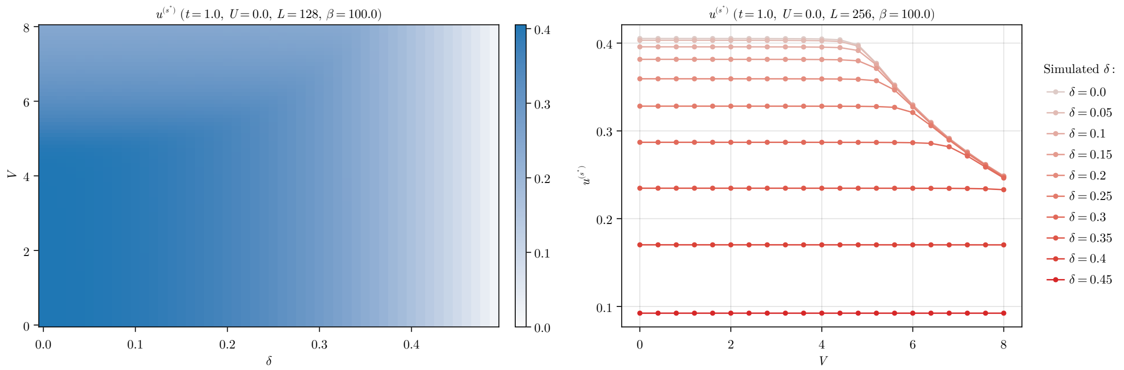
Following this line of reasoning, we can extract easily the two limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta,0]\big|_{\beta\gg 1} \simeq \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 1.3:

$$\begin{aligned}
u^{(s^*)}[\beta,0]\big|_{\beta\gg 1} &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\cos k_y \\
&= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\cos k_x \\
&= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\sin(\pi-k_y) \\
&= \left(\frac{2}{\pi}\right)^2 \approx 0.405
\end{aligned} \tag{1.27}$$



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 1.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2. At unitary filling we have

$$u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right]\Big|_{\beta\gg 1} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in \mathrm{BZ}}(\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarize: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.

1.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs "vector" defined in Sec. **??** has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterized by the first self-consistency equation of Tab. 1.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$

$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

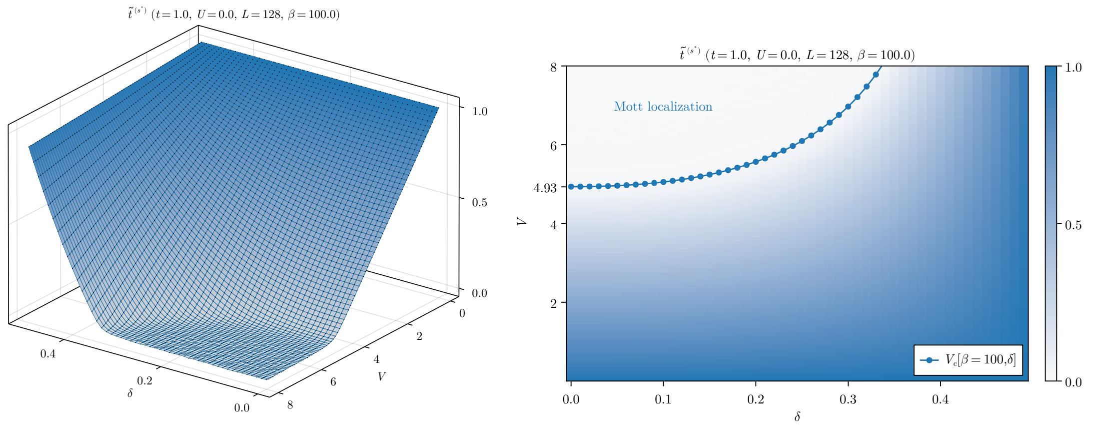
the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (1.22). Evidently the normal phase is completely independent of U, since the local repulsion cannot develop any RWC other than those renormalizing the chemical potential. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, setting up the following general parameters:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)}: 5\times 10^{-4}\right]$$

with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. **??**.

In Fig. 1.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$



a | Simulated values of $\tilde{t}^{(s^*)}$ b | Critical Mott boundary.

Figure 1.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (1.28). For graphic reasons the points at $V_c > 8$ have been omitted.

The expectations of Sec. 1.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 1.6a. This effect is more visible in Fig. 1.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 1.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (1.27).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 1.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localization effect.

1.6.1 Mott localization effect

As is evident from Fig. 1.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$\forall V \mid 0 \leq V < V_c[\beta, \delta] \quad \Longrightarrow \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

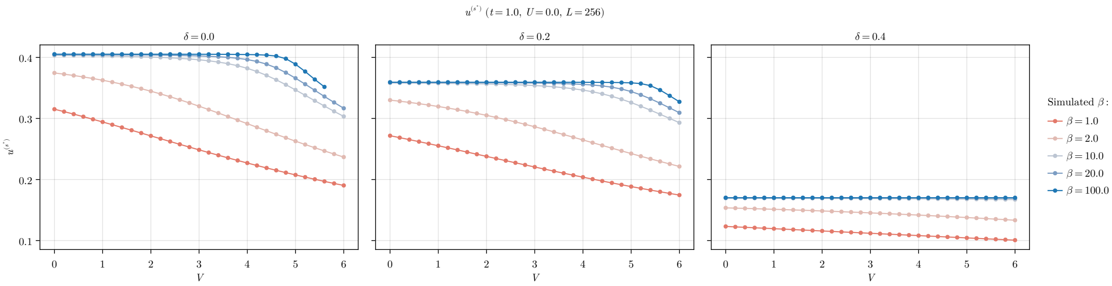
Then within the entire segment $[0, V_c]$ the convergence value is the same obtained for $V = 0$, hence we can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}}$$

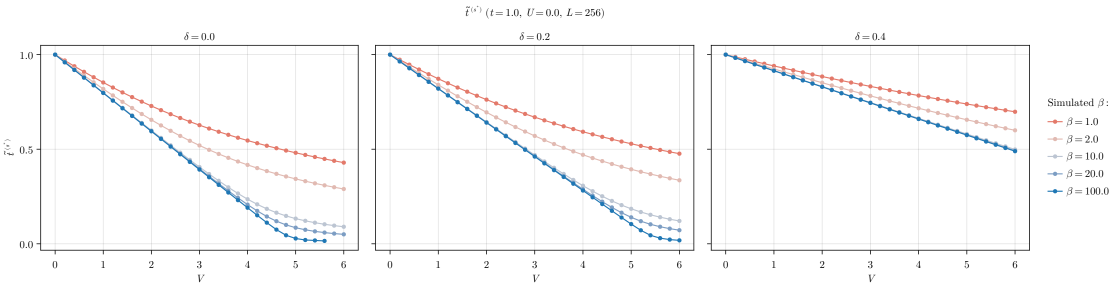
When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (1.24) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (1.24) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 1.7a the converged values of $\tilde{t}^{(s^*)}$, extracted immediately from the same data of Fig. 1.6a, are plotted in parametric space.

While to extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature, for the high-β half-filled case it is immediate from Eq. (1.27) to see that

$$V_c[\beta \gg 1, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{1.28}$$



a | Temperature dependence of $u^{(s^*)}$ plots at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ plots at various dopings.

Figure 1.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localization, reducing the bands shrinking effect.

which makes the real interesting point of this section and the whole bands-shrinking effect. At half-filling, the effective bands amplitude reduction is so strong that already around relatively small values of V (compared to t) electronic kinetics is nullified. In Fig. 1.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure described. As is evident from the plot itself, even if a little rough, our prediction of the critical value for V gives reasonably a sort of phase boundary delimiting a region where bands are flattened and electrons are localized.

This phenomenon is largely studied in solid-state physics and falls within the large class of Mott physics. [Blabla about Mott physics...]

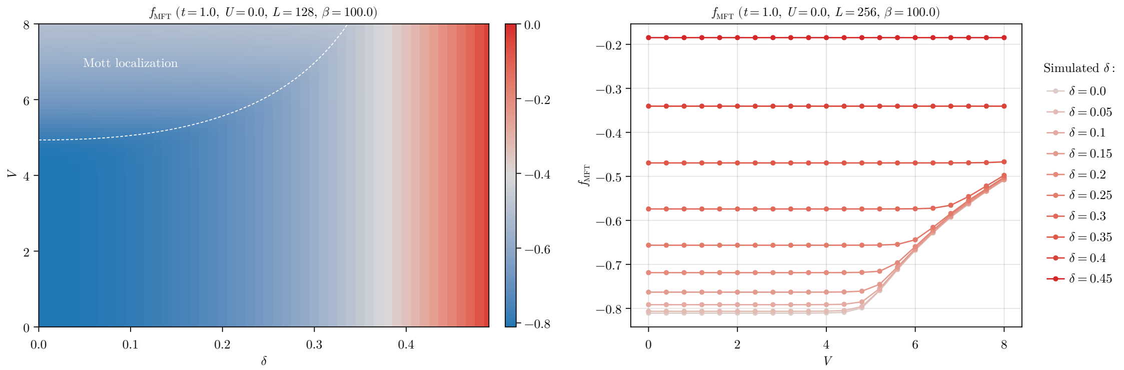
Increasing temperature, predictably, Mott localization is suppressed. In Fig. 1.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$\beta = 1.0, 2.0, 10.0, 20.0, 100.0 \qquad \delta = 0.0, 0.2, 0.4$$

[Evaluate inserting more temperatures.] At high temperature the V-independence region completely disappears, making $u^{(s^*)}$ a monotonically descendent function. This is no surprise, since all our derivation was performed at low temperature. In Fig. 1.8a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localization effects, at least at small values of V. Interestingly enough, however, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

1.6.2 Solution free energy and entropy densities

The plots of Fig. 1.9 report our findings on the Normal phase free energy, computed based on Eq. (1.22). Doping induced energy increase is simply due to the simultaneous entropy decrease and band filling (the first being a consequence of the second, because less and less room is available for electron scattering). By looking the data of Fig. 1.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (1.22) compensates the free energy increase



a | Heatmap of f_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 1.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition.

due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}})\, f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}
$$

having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V (\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. 1.4, to add said correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

From Fig. 1.9b, energy cost of Mott localization is made evident. The lower the doping, the higher the energetic cost for total bands shrinking. At low doping levels, hopping renormalization is the largest, as discussed in previous sections; at low temperatures the band is full below $\tilde{\mu}$, thus by shrinking it we just "move up" single-state energies, but as long as a FS is well-defined (which is, as long as the band is not flattened enough) the Fermi sea is essentially the same. Including the aforementioned correction, correctly, gives a total free energy density that is indistinguishable from the one of the Normal phase for the HM. On a flat band a FS is not well defined, and because of this energy must rise: in the Mott-localized region the Fermi-Dirac factors of Eq. (1.22) degenerate to

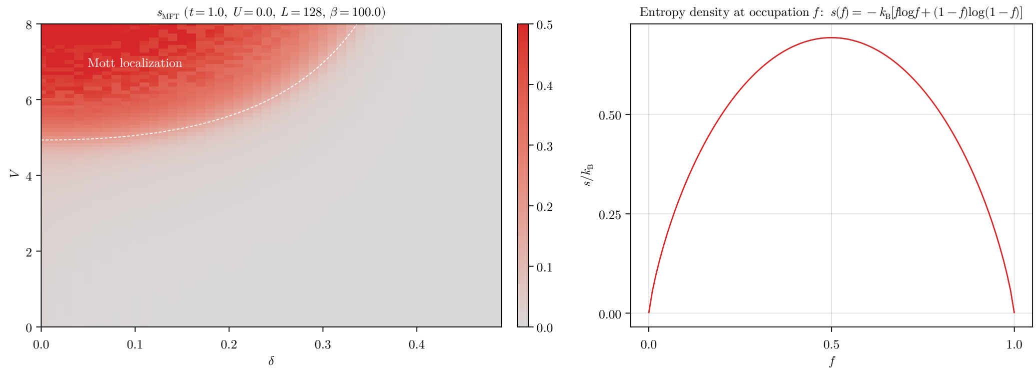
$$
\text{Mott localized} \quad \Longrightarrow \quad \lim_{\beta\to\infty} f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}
$$

since for the chemical potential $\tilde{\mu}(\delta) = 0$. From a free, single band Fermi gas the contribution to entropy given by a state with occupation f is [8]

$$
s(f) = -k_{\mathrm{B}} \left[f \log f + (1 - f) \log(1 - f) \right]
$$

represented in Fig. 1.10b. The maximum is precisely located at $f = 1/2$, which from the limit above we know is the limiting expression for the Fermi-Dirac population of each single-particle state for a flat band. This implies that the key free energy non-trivial contribution is entropic, represented in Fig. 1.10a. Correctly entropy is low all around the Mott insulating phase, which is coherent with a normal Fermi gas at low temperature, for which the FS interface (the region where the Fermi-Dirac factor varies continuously from ~ 1, inside, to ~ 0, outside) is narrow. For said model almost all single-particle state are approximately full or empty, which as is represented in Fig. 1.10b are two symmetric low-entropy conditions. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.



a | Heatmap of s_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 1.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition. [Perhaps use a lighter gray in heatmap.]

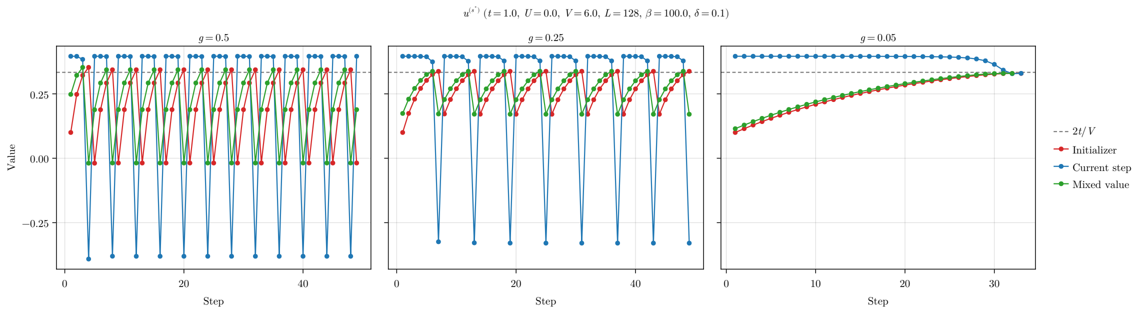


Figure 1.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

1.6.3 Convergence problems and mixing fine-tuning

Although extremely simple and characterised by a single HFP, this framework is ideal for us to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. **??**. Bands flattening is a source of algorithmic instability: numeric fluctuations can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (1.24). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract converged values, it is necessary to pay a little care with the choice of the mixing parameter $g \in [0, 1]$. We defined it in Sec. **??**, and operatively in Eq. (**??**). This parameter controls deterministically how, from an initializer (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initializer for the following step is generated. As a general rule, it is pretty safe to use $g \approx 0.5$; this rule of thumb, however, breaks in situations like the one plotted comparatively in Fig. 1.11, representing the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

As can be checked in Fig. 1.6a this is a point well inside the Mott region. Localization, as we see in Fig. 1.7, is already pretty intense and to sustain bands flattening $u^{(s^*)}$ needs to converge at $2t/V = t/3$. Let us follow the evolution of the algorithm starting from the arbitrary initializer

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

In each panel of Fig. 1.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue one the result, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target. In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. As we know the generated green point is out of bounds with respect to Eq. (1.24). When this happens, the next red point (initializer of following step) gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence, implementing the idea that – within a given tolerance – even if the blue line is systematically above the convergence value (in the region of band sign inversion) our mixing can be made careful enough to make up for this, and generate a green track systematically below the convergence value. This kind of problems can turn out to be pretty common for iterative HF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 2

Antiferromagnetic instability

This chapter is devoted to an in-depth discussion of the antiferromagnetic (AF) long-range ordering in the EHM by the means of MFT. In the context of the 2D HM, discussed in App. A, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. A.1.8). The situation in EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the gap "complexification" and the bands renormalization and resymmetrization (both in Sec. 2.3), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) translational invariance, discussed previously in Sec. 1.3.

2.1 The AF phase and its symmetries

To start, AF phase is not uniquely defined; actually, there exists a variety of ways to establish a SDW phase in a spin-balanced square lattice, a system where both spin sectors are equally populated, but differently displaced. Fig. 2.1 presents three (of many) types of antiferromagnets. The left panel shows the one we deal with: SDW structures are 2-sites periodic, making the AF phase "commensurate". The central one presents another simple commensurate AF with higher periodicity. The right panel shows a possible striped antiferromagnet, where AF is effectively established along one particular direction. [Add: striped AF is a studied phase, blabla. Some connection with nematicity.]

The AF phase hereby presented is a symmetry reduction of the normal phase, and is described

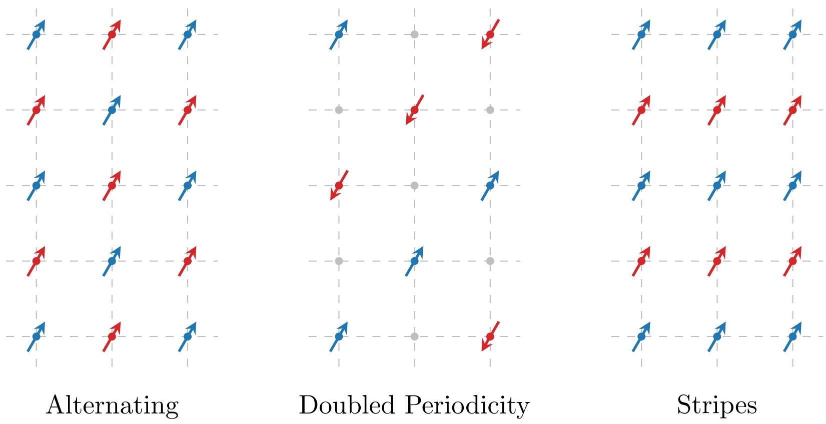


Figure 2.1 Three possible antiferromagnetic structures. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

Operator	**Sector**	**RWCs**	**Net effect**
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization, gap complexification

Table 2.1 Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. 1.1.

by simultaneous spin rotational symmetry reduction and translational symmetry halving:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$$
$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$$

We describe a SDW whose period is of two sites, as in the left panel of Fig. 2.1. As done for the normal phase, we allow theoretically

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

in order to capture the same possible nematic effects seen in Sec. 1.3.1.

Finally, recall Tabs. **??** and **??**, collecting the RWCs for the AF phase,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the role of all RWCs in Tab. 2.1.

2.1.1 Antiferromagnetism in the HM

In this short section we recall the principal features of the AF phase established in the conventional HM, discussed in Sec. A.1. Before starting, let us define as in Eqns. (A.3) and (A.2) the AF and density fluctuations operators,

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{2.1}$$
$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{2.2}$$

which represent the two essential fermionic quadratic operators of the analytic MFT derivation. It is very useful to also note the following properties. Given the γ-wave form factors of Tab. **??**, evidently both the following equations hold

$$\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = \hat{\psi}_{\mathbf{k}\sigma}$$
$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}}$$

which imply the handy property

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\psi^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\psi_{\mathbf{k}\sigma} \tag{2.3}$$

Within the pure HM, the AF phase is specified by the Ansatz (A.4),

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{2.4}$$

which is explicitly breaking translational invariance in each spin sector, while preserving $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. As reported in Tab. **??**, the only RWC comes in Hartree channel, since the

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.20)
$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$i \left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.21)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$	(A.22)

Table 2.2 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

local two-body interaction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (A.5)

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] \\ - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\text{AF})}_{\text{H}/U}$$

with $E^{(\text{AF})}_{\text{H}/U}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (A.6),

$$E^{(\text{AF})}_{\text{H}/U} = L_x L_y \times U \left(n^2 - m^2 \right)$$

In reciprocal space, the hamiltonian decomposes as in Eq. (A.11),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU \times \hat{N} - E^{(\text{AF})}_{\text{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and the free electrons energy is simply the tight binding energy of Eq. (??) written with an explicit spin index,

$$\epsilon_{\mathbf{k}\sigma} = -2t \left(\cos k_x + \cos k_y \right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (A.24),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}$$

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 2.2. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 2.2.

2.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$. Being the RWCs the same as for the normal phase, the discussion is essentially the same of Chap. 1, with the only relevant difference being the Ansatz we use in Eq. (1.1), substituted by the commensurate oscillatory Ansatz of Eq. (2.4). By itself, even if it may seem the contrary, the Ansatz does not imply C_4 symmetry preservation. It is possible to reduce point group symmetry having still a rotation-symmetric magnetization, as well as it was possible to do in the normal phase with keeping a uniform density. As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.5}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.6}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.7}$$

equivalent to the ones for the pure HM, Eqns. (A.20), (A.21) and (A.22). Moving to the the tilted frame, the z component of the pseudospin is just a thermal superposition of the "up" and "down" states

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{2.8}$$

Let us define the (renormalized) thermal factors as in Eq. (A.19),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{2.9}$$

The angles satisfy:

$$\sin 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{2.10}$$

The physical behaviour is the same as for the pure HM. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. 1: the RWCs are the same of Sec. 1.1, listed in Tabs. **??** and **??**. All three Hartree contractions, responsible in the normal phase for the chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start by the Hartree terms: due to the "staggered" nature of the commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing the MFT approximations effect on bands. Next sections are basically an extension of the calculations of Chap. 1.

2.2 Non-local Hartree effects: chemical potential renormalization

As in Eq. (1.3), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right]}^{\text{s.s.}}$$
$$\underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. We use Eq. (2.4), which describes an oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n, and the oscillatory part

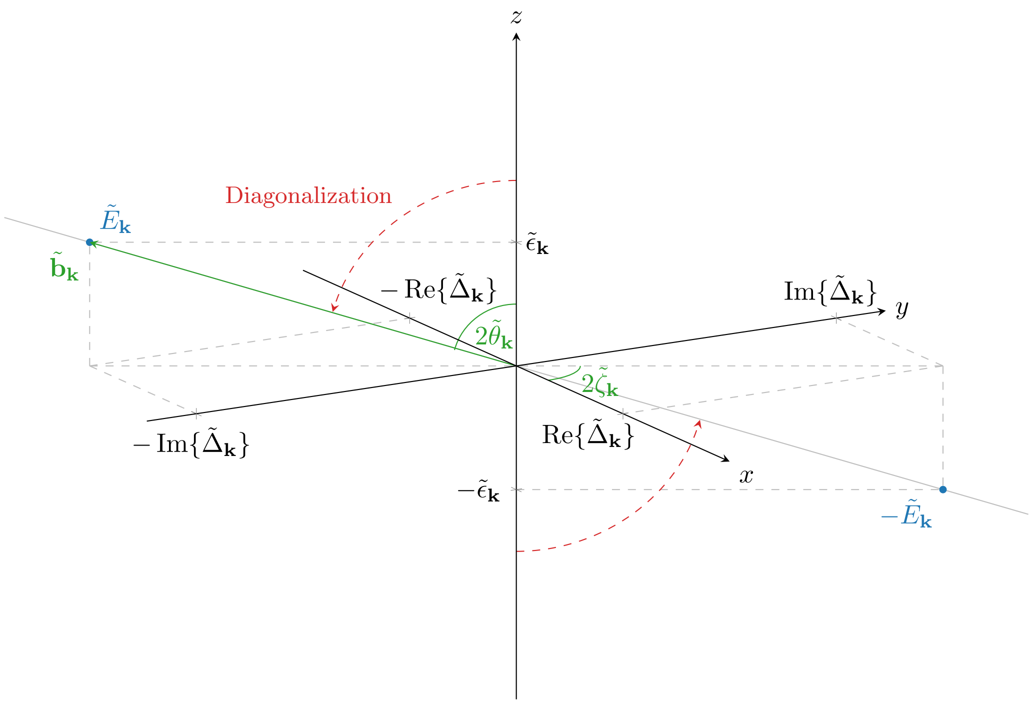


Figure 2.2 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma = \uparrow$ has been omitted.

m. The algebraic derivation relative to the n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (1.3),

$$-V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\sigma}\rangle\right]-V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\overline{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle\right] \tag{2.11}$$

as well as the its scalar counterpart

$$V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle+V\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle \tag{2.12}$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

2.2.1 Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (2.11). By inserting the Ansatz of Eq. (A.4), we get

$$\overbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}(-1)^{\delta_{\sigma=\uparrow}}\left[(-1)^{x+y}\hat{n}_{\mathbf{r}'\sigma}+(-1)^{x'+y'}\hat{n}_{\mathbf{r}\sigma}\right]}^{\text{s.s.}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}'\overline{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\right]+mV\sum_{\langle\mathbf{rr}'\rangle}\sum_{\sigma}\left[(-1)^{x+y+\delta_{\sigma=\uparrow}}\hat{n}_{\mathbf{r}'\overline{\sigma}}+(-1)^{x'+y'+\delta_{\overline{\sigma}=\uparrow}}\hat{n}_{\mathbf{r}\sigma}\right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r}=(x,y)$ and $\mathbf{r}'=(x',y')$ are NNs, evidently

$$(-1)^{x'+y'}=(-1)^{x+y+1}$$
$$(-1)^{\delta_{\overline{\sigma}=\uparrow}}=(-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (density)}}\overbrace{-mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}\right]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}\right]}_{\text{o.s. (density)}}+\underbrace{mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}\right]}_{\text{o.s. (magnetization)}} \tag{2.13}$$

In the above expressions the various contribution have been separated in “density” contributions and “magnetization” contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (2.13),

$$-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}\right]-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}\right] \tag{2.14}$$

The discussion is identical to the one of Sec. 1.2.1 for the normal phase, with the renormalization of μ being the same of Eq. (1.5).

$$\tilde{\mu}\equiv\mu+n(2zV-U) \tag{2.15}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$., discussed both in Sec. 1.2 and Sec. A.1.1.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (2.13) perfectly cancel out:

$$mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}})\right]=0$$

thus the only net effect of the Hartree RWCs affects μ. This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

2.2.2 Hartree contraction energy

Consider the scalar Expr. (2.12): it’s the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contractyion energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the saeme to happen for double counting terms.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle=\overbrace{V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow}$$
$$+\underbrace{V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboiring site, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 - m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n + (-1)^{x+y} m)(n - (-1)^{x'+y'} m) \right]}^{\sigma=\uparrow} + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n - (-1)^{x+y} m)(n + (-1)^{x'+y'} m) \right]}_{\sigma=\downarrow}$$

which gives immediately

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\overline{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 + m^2)$$

Hence, the total "contraction energy", given by the sum of both contributions, is identical to the normal one of Eq. (1.7) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 = 4n^2 V \times \frac{z}{2} L_x L_y \tag{2.16}$$

the minus sign coming from the fact that we are subtracting this energy shift.

2.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. 1.3 derivation, up to Eq. (1.9),

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{2.17}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (1.10), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

2.3.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$\left| \langle \hat{n}_{\mathbf{k}\sigma} \rangle \right| \geq 0 \qquad \left| \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

Because of this, for the conventional HM the AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (A.30). Assuming the new ground state to be similar and going back to Eq. (2.17), this implies only two contributions are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{n}_{\mathbf{k}\sigma} \rangle$, the second the staggered AF order parameter, $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle$. Then Eq. (2.17) is reduced to:

$$(\cdots) \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \Big[\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \Big] \tag{2.18}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (2.18) are simple density interactions with the mean density field. Consider Fig. 1.1a: density at vector $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. These variables are depicted in Fig. 1.1b. Apply in the diagonal part of Eq. (2.18) this variables change (which is the same of Eq. (1.13))

$$\begin{aligned}
&\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&\qquad = \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\
&\qquad = \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma}
\end{aligned} \tag{2.19}$$

where in the last passage we substituted the decomposition of Eq. (**??**). Define as in Eq. (1.15) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \tag{2.20}$$

The expression in square brackets of Eq. (2.19) is just its γ component, and is expanded in

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{2.21}$$

where, in the last passage, Eqns. (2.7), (2.8) and (2.9) have been used. Now, the AF phase we realize does not break inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalized bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_{\sigma} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 2.3 using spin degeneracy.

Recognizing the harmonics, Eq. (2.19) becomes

$$\begin{aligned}
&\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&\qquad = V \sum_{\mathbf{q},\sigma} \left[u^{(s^*)}_{\sigma} (\cos q_x + \cos q_y) + u^{(d)}_{\sigma} (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}
\end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. **??**

Table 2.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{2.22}$$

as was for the normal phase in Eq. (1.17). In Sec. 1.3.3 we discussed the appearance of anisotropic hopping with d-wave extension of the bands: here the theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (2.18). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 1.1b,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \qquad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (2.19)

$$\begin{aligned} &-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\ &\qquad = -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}'\sigma} \end{aligned}$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k}+\boldsymbol{\pi} \equiv \mathbf{q}$, and use Eq. (2.3) as well as reciprocal space periodicity by a BZ,

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

which finally gives

$$\begin{aligned} &-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\ &\qquad = \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}^\dagger_{\mathbf{q}'\sigma} \end{aligned} \tag{2.23}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \tag{2.24}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}(\cos k_x + \cos k_y)\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}^{(-)}_\mathbf{k}[\beta,\tilde{\mu}]$	Fig. **??**
p_x-wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}\sin k_x\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}^{(-)}_\mathbf{k}[\beta,\tilde{\mu}]$	Fig. **??**
p_y-wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}\sin k_y\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}^{(-)}_\mathbf{k}[\beta,\tilde{\mu}]$	Fig. **??**
$d_{x^2-y^2}$-wave	$v^{(d)} = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\text{MBZ}}(\cos k_x - \cos k_y)\frac{\text{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tilde{\text{Th}}^{(-)}_\mathbf{k}[\beta,\tilde{\mu}]$	Fig. **??**

Table 2.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_\mathbf{k}$.

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned} v_\sigma^{(\gamma)} &= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\text{BZ}}\varphi_\mathbf{q}^{(\gamma)}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle \\ &= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\text{MBZ}}\left[\varphi_\mathbf{q}^{(\gamma)}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)}\langle\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right] \\ &= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\text{MBZ}}\varphi_\mathbf{q}^{(\gamma)}\left[\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle - \langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\right] \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\text{MBZ}}\varphi_\mathbf{q}^{(\gamma)}\langle\hat{\Psi}^\dagger_{\mathbf{q}\sigma}\tau^y\hat{\Psi}_{\mathbf{q}\sigma}\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\text{MBZ}}\varphi_\mathbf{q}^{(\gamma)}\frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\text{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \end{aligned} \tag{2.25}$$

where we made use of Eq. (2.3), the relations of Tab. 2.2 as well as Eq. (2.6). There is no particular reason to exclude one component or another. In Tab. 2.4 the four derived self-consistency equations are reported (already ignoring spin index).

From the definitions of the structure factors of Tab. **??**, it's immediate to see that:

$$v_\sigma^{(s^*)}, v_\sigma^{(d)} \in \mathbb{R} \qquad v_\sigma^{(p_x)}, v_\sigma^{(p_y)} \in i\mathbb{R}$$

Getting back to Eq. (2.23) and inserting these components, we get

$$\begin{aligned} &-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\ &\qquad = -iV\sum_{\gamma,\sigma}v_\sigma^{(\gamma)}\sum_{\mathbf{q}\in\text{BZ}}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}\sigma} \end{aligned} \tag{2.26}$$

We can expand on the MBZ as

$$\begin{aligned} &-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\ &\qquad = -iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\text{MBZ}}\left[v_\sigma^{(\gamma)}\varphi_\mathbf{q}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}\sigma} + v_\sigma^{(\gamma)}\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)*}\hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\right] \end{aligned}$$

and use Eq. (2.3), exchanging the terms positions

$$-\frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}-v^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^{\dagger}_{\mathbf{q}\sigma}\right] \quad (2.27)$$

Now: recalling Tab. ??, for the four symmetries we are considering, the form factors are real for symmetries s^* and d, and purely imaginary for p_x and p_y. As we mentioned before, the same holds for the $v^{(\gamma)}_{\sigma}$ harmonics. Then

$$iv^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}\in i\mathbb{R} \quad (2.28)$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-iv^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}=\left(iv^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}\right)^{*}$$

Therefore, we conclude

$$-\frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+\mathrm{h.c.}\right] \quad (2.29)$$

which is obvious, being the hamiltonian operator hermitian from the beginning.

2.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (2.17). As said before, its definition is the same as for the normal phase, Eq. (1.10), with the averaging operation being performed on the AF ground state:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}\equiv\frac{V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}[\cos(\delta k_x)+\cos(\delta k_y)]\,\langle\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \quad (2.30)$$

Proceeding exactly as in Eq. (2.18) and making the change of variables of Fig. 1.1, we get

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}=\frac{V}{L_x L_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}[\cos(\delta q_x)+\cos(\delta q_y)]\Big[\underbrace{\langle\hat{n}_{\mathbf{q}\sigma}\rangle\langle\hat{n}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}}-\underbrace{\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\langle\hat{\psi}_{\mathbf{q}'\sigma}\rangle}_{\text{Off-diagonal terms}}\Big]$$

Apply the decomposition of Eq. (??) and use Eqns. (2.20), (2.24),

$$E^{(\mathrm{AF})}_{\mathrm{F}/V}=\frac{V}{2L_x L_y}\sum_{\gamma,\sigma}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\left[u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}+v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma}\right] \quad (2.31)$$

Now: it holds

$$u_{\mathbf{q}\sigma}=\langle\hat{n}_{\mathbf{q}\sigma}\rangle \quad\Longrightarrow\quad u_{\mathbf{q}\sigma}=u^{*}_{\mathbf{q}\sigma}$$

and then

$$\frac{1}{(L_x L_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}=\left[\frac{1}{L_x L_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_x L_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}u^{*}_{\mathbf{q}'\sigma}\right]$$
$$=\left|u^{(\gamma)}_{\sigma}\right|^{2}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}=\hat{\psi}^{\dagger}_{\mathbf{q}\sigma}$,

$$v_{\mathbf{q}\sigma}=i\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle \quad\Longrightarrow\quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma}=-v^{*}_{\mathbf{q}\sigma}$$

Since the form factors satisfy $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$, we get:

$$\begin{aligned}
\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}v_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}'\sigma}\right]\\
&= v^{(\gamma)}_{\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}+\boldsymbol{\pi}}v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]\\
&= v^{(\gamma)}_{\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}}v^{*}_{\mathbf{k}\sigma}\right]\\
&= \left|v^{(\gamma)}_{\sigma}\right|^2
\end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k}\in$ BZ. Inserting the above results in Eq. (2.31), this last becomes:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = L_xL_y\times\frac{V}{2}\sum_{\gamma,\sigma}\left[\left|u^{(\gamma)}_{\sigma}\right|^2+\left|v^{(\gamma)}_{\sigma}\right|^2\right] \tag{2.32}$$

Interestingly enough, the new HFPs amplitudes contribute positively to the contraction energy. This fact is non-trivial and has consequences we will discuss in Sec. 2.6.1.

2.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned}
\hat{H}-\mu\hat{N} \simeq \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\tilde{\epsilon}_{\mathbf{k}\sigma}\left[\hat{n}_{\mathbf{k}\sigma}-\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\tilde{\Delta}_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma}+\tilde{\Delta}^{*}_{\mathbf{k}\sigma}\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]\\
-\tilde{\mu}\hat{N} - \left[E^{(\mathrm{AF})}_{\mathrm{H}/U}+E^{(\mathrm{AF})}_{\mathrm{H}/V}+E^{(\mathrm{AF})}_{\mathrm{F}/V}\right]
\end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V\sum_{\gamma\in\{s^*,d\}}u^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{k}}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV\sum_{\gamma}v^{(\gamma)}_{\sigma}\varphi^{(\gamma)*}_{\mathbf{k}} \tag{2.33}$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}}mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle\hat{s}^{x}_{\mathbf{k}\sigma}\rangle = -\langle\hat{s}^{x}_{\mathbf{k}\overline{\sigma}}\rangle$$

which is relevant in the computation of the physical magnetization, which (as is proven in Eq. (A.10)) is defined

$$\begin{aligned}
m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\rangle\\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\langle\hat{s}^{x}_{\mathbf{k}\uparrow}-\hat{s}^{x}_{\mathbf{k}\downarrow}\rangle\\
&= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\langle\hat{s}^{x}_{\mathbf{k}\uparrow}\rangle
\end{aligned}$$

A proof of the above derivation is given in Sec. A.1.7. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left|\operatorname{Im}\{\Delta_{\mathbf{k}\sigma}\}\right|\geq 0 \quad\Longrightarrow\quad \left|\langle s^{y}_{\mathbf{k}\sigma}\rangle\right|\geq 0$$

while for the HM the y pseudospin component was identically null.

2.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. A.1.4, we derive analytically the AF ground state on the conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudospins collective pure state. Eq. (A.29) reports the final form of said ground-state. Said quantum state is in general pretty similar to the conventional BCS state.

At half-filling and zero temperature, it is rather easy to extend that state to the new complexified gap. By performing an identical calculation as in Sec. A.1.4 we here omit, it's easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs in the various amplitudes and a relative phase taking care of the gap function complex phase,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\text{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{2.34}$$

which is identical to Eq. (A.29), apart from renormalizations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\text{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma}\right]|\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}^\dagger_{\mathbf{k}\sigma}\}|\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in the above equations are the same of Eq. (2.10), which acquire a physical meaning: the relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned}
\langle\Psi|\hat{s}^x_{\mathbf{k}\sigma}|\Psi\rangle &= \langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Psi\rangle \\
&= \sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\cos 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

$$\begin{aligned}
\langle\Psi|\hat{s}^y_{\mathbf{k}\sigma}|\Psi\rangle &= i\,\langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Psi\rangle \\
&= i\sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\
&= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\sin 2\tilde{\zeta}_{\mathbf{k}\sigma}
\end{aligned}$$

These equations are coherent with Eqns. (2.5), (2.6), thus the involved angles apart from irrelevant 2π shifts coincide.

2.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying is not allowed to break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \text{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \text{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \text{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\text{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs listed in Tab. 2.4: s^* and d-wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 2.5 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

sectors. We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 2.5. The presence itself of two "AF flavours" is not-so-physical, given that up to now we have no clear physical interpretation for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. 2.7, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not inferrable from MFT analysis, but can be seen only delving into the detailed discussion of the iterative algorithm seen as a recursive parametric map. For now, let us proceed integrally with all the HFPs we found and derive the results.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. 2.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it's relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{2.35}$$

which is the actual physical magnetization. From the first line we see that it can be interpreted as the average transverse pseudospin value.

2.5 Stability of the AF phase

[Re-write and move up.]

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. ??. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [15]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. ?? and mimicking Eq. (A.34), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\, \chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{2.36}$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [17], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;

5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

Let us gather some more informations on the HFPs we identified.[To be continued...]

2.6 AF free energy density

In order to derive the expression of the free energy density for the AF phase, we proceed precisely as in Sec. 1.4. The only notable difference is in entropy: Eq. (1.21) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \Big] \tag{2.37}$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \Big] + E^{(\mathrm{AF})}_{\mathrm{c}}$$

being $E^{(\mathrm{AF})}_{\mathrm{c}}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. 1.4, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e^{(\mathrm{AF})}_{\mathrm{g}} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:

$$e^{(\mathrm{AF})}_{\mathrm{c}} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2 \right) \right]$$

3. Entropic energy

$$e^{(\mathrm{AF})}_{s} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. Chemical potential correction

$$e^{(\mathrm{AF})}_{n} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_{s}^{(\mathrm{AF})} + e_{n}^{(\mathrm{AF})}$$

and we get

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] \\ + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + n\left[\mu + n(2zV - U)\right]$$

As was for the normal phase, the contraction energy and the chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (2.15),

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] \\ + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + \tilde{\mu}n \quad (2.38)$$

2.6.1 The gap energy

It is useful to see here a property of the above expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V\sum_{\gamma}\left|v^{(\gamma)}\right|^2 = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \quad (2.39)$$

The proof is simple and immediate: developing the right-hand side,

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ + \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]$$

Now:

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned}\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= Um \times \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ &= Um^2\end{aligned}$$

having we recognized the magnetization from Eq. (2.35). Equivalently,

$$\begin{aligned}\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V\sum_{\gamma} v^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V\sum_{\gamma} v^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

where we used Eq. (2.28) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned}\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= -V\sum_{\gamma} v^{(\gamma)*}\sum_{\mathbf{k}\in\mathrm{MBZ}}\varphi_{\mathbf{k}}^{(\gamma)}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ &= -V\sum_{\gamma} v^{(\gamma)*} \times v^{(\gamma)} \\ &= -V\sum_{\gamma}\left|v^{(\gamma)*}\right|^2\end{aligned}$$

where in second passage we recognized $v^{(\gamma)}$ given in Eq. (2.25). Then Eq. (2.39) is verified. The latter is particularly interesting because it shows a non-intuitive result: due to gap complexification, gap amplitude is only increased by $\hat{H}_V$, thus by analogy we would expect the relative free energy contribution to have the same sign as the original gap energy from the real part. This is not the case: to open a gap in the s-wave real sector contributes positively to free energy, while to open a gap in the γ-wave imaginary sector contributes negatively.

An immediate consequence of Eq. (2.39) is that, being the right-hand side evidently positive, for $V > 0$ the total amplitude of the "complexifiers" vector is maximised,

$$\sum_{\gamma} \left|v^{(\gamma)}\right|^2 \leq M \quad \text{being} \quad M^{(\mathrm{AF})} \equiv m^2 \frac{U}{V} \tag{2.40}$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- Correctly, when the physical magnetization closes, so does the imaginary part of the gap

  $$\lim_{m \to 0} v^{(\gamma)} = 0$$

  which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (2.39) is also solved by a null magnetization and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

[Graphic interpretation: both the aAF and sAF phases have two-syms vs, thus the constraint above tells us that the relative two-elements vector is contained in a circumference.]

2.6.2 Free energy at half-filling

It's of particular interest the half-filling condition, with $\tilde{\mu} = 0$, where PHS discussed in Sec. **??** is preserved in the AF phase. Since, as it can be shown easily,

$$\begin{aligned} \ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x) \end{aligned}$$

Let now $x = \beta \tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \to +\infty$, which gives

$$x - 2\ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \tilde{E}_{\mathbf{k}} + U m^2 - V \sum_{\gamma} \left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right]$$

Starting from the full expression of the free energy density, we recovered the intuitive expression of the energy of a fully (double) occupied band at $-\tilde{E}_{\mathbf{k}}$, with HFPs correction.

[Expand: interesting situation for bands with nodes. At low temperature, where band nodes are present, the function behaves differently because its argument goes to zero. What is the behaviour there?]

2.7 Theoretical constraints on the AF phase HFPs

Up to now, all we derived was the most general MFT-compatible mathematical description of a commensurate AF established on the EHM, allowing for nematic spatial symmetry breaking as well as non-trivial renormalization effects. This section, a clone of Sec. 1.5, is dedicated to understanding, given the algorithmic problem, how its solutions should behave. As we will see, based on this theoretical analysis we are able to reduce a lot the algorithmic complexity and limit the solution search to a tailored subspace of HFPs space.

We start from considering a robust C_4 antiferromagnet, thus setting $u^{(d)} = 0$ by hand, and show how the constraints on $u^{(s^*)}$ mimic those posed for the normal phase in Eq. (1.24). Then we move to a rather convoluted but necessary analysis of the five components HFPs vector described for both AF flavours in Tab. 2.5. For the sake of generality, even if – as discussed in Sec. 2.5 – at finite doping the system is naturally unstable, we take on all calculations at generic $\tilde{\mu}$.

2.7.1 Robust C_4 bands shrinking extension to AF phases

For the normal phase, we saw that hen C_4 symmetry is preserved (Sec. 1.5.1) we can conclude immediately the bands renormalization is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$
the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives
$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$
But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily
$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$
the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives
$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$
We conclude necessarily
$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V \tag{2.41}$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

2.7.2 HFPs vanishing in sAF phase

We now present a somewhat convoluted argument to show that, even if raw MFT presents an apparently rich scenario with a lot of HFPs, the problem can be proven to be actually much simpler. Let us start from the sAF phase, assuming

$$C_4 \overset{\mathrm{SSB}}{\rightarrow} C_2$$

The system is described by the five HFPs on the first row of Tab. 2.5 and their self-consistency equations of Tabs. 2.3 and 2.4.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \tag{2.42}$$

whose linearity properties are trivial,

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \tag{2.43}$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \Longrightarrow \quad F[f(\mathbf{k})] \geq 0 \tag{2.44}$$

The relation above is implied by the fact that both $\tilde{\text{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\text{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}}-\mu)}+1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}}-\mu)}+1} \geq 0 \qquad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ &= \frac{\left[2t - Vu^{(s^*)}\right]}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)} V\right](c_x + c_y) + u_\sigma^{(d)} V (c_x - c_y)$$

Evidently, in terms of the functional S, we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F\left[(c_x + c_y)^2\right] - Vu^{(d)} \times F\left[c_x^2 - c_y^2\right] \\ &= \left[2t - Vu^{(s^*)}\right] \times \left\{F\left[c_x^2\right] + 2F\left[c_x c_y\right] + F\left[c_y^2\right]\right\} - Vu^{(d)} \times \left\{F\left[c_x^2\right] - F\left[c_y^2\right]\right\} \end{aligned}$$

having we used the linearity property of Eq. (2.43) in last passage.

Let now:

$$A \equiv F\left[c_x^2\right] \qquad B \equiv 2F\left[c_x c_y\right] \qquad C \equiv F\left[c_y^2\right] \tag{2.45}$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C) \tag{2.46}$$

$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A - C) - Vu^{(d)}(A - B + C) \tag{2.47}$$

$$v^{(s^*)} = -Vv^{(s^*)}(A + B + C) - Vv^{(d)}(A - C) \tag{2.48}$$

$$v^{(d)} = -Vv^{(s^*)}(A - C) - Vv^{(d)}(A - B + C) \tag{2.49}$$

These equations define a four-fold nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a conformal transform on the map: define the new "sum" and "difference" variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \qquad D_u \equiv u^{(s^*)} - u^{(d)} \qquad S_v \equiv v^{(s^*)} + v^{(d)} \qquad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (2.46) and (2.47) (the first two) as well as Eq. (2.48) and (2.49) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B) \tag{2.50}$$
$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C) \tag{2.51}$$
$$(1 + 2AV)S_v + BVD_v = 0 \tag{2.52}$$
$$(1 + 2CV)D_v + BVS_v = 0 \tag{2.53}$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (2.53), using $C = F\left[c_y^2\right] \geq 0$ implied by Eq. (2.44):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \qquad \Longrightarrow \qquad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (2.52), $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (2.52): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V} \frac{S_v}{D_v}$$

which is non-zero because $A = F\left[c_x^2\right] \geq 0$, the latter following from Eq. (2.44). We could have used identically Eq. (2.53).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV) \tag{2.54}$$

To prove the latter, it is sufficient to multiply Eq. (2.52) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (2.53) $BVS_v = -(1 + 2CV)D_v$,

$$\left[-(1 + 2CV)(1 + 2AV) + (BV)^2\right] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (2.54) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \tag{2.55}$$

To prove this: multiply Eq. (2.50) by BV on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B)$$

and substitute from Eq. (2.51) $BVS_v = -(1 + 2CV)D_v + 2t(B + 2C)$,

$$\begin{aligned} -(1 + 2CV)(1 + 2AV)D_v + +2t(B + 2C)(1 + 2AV) + (BV)^2 D_u &= 2tBV(2A + B) \\ 2t(B + 2C)(1 + 2AV) &= 2tBV(2A + B) \\ B + 2ABV + 2C + 4ACV &= 2ABV + B^2 V \end{aligned}$$

Passing from first to second line we used Eq. (2.54), and in the passage below $t > 0$. Last line implies Eq. (2.55).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (2.54),

$$(BV)^2 = 1 + 2V(A + C) + 4ACV^2 \quad \Longrightarrow \quad 1 + 2V(A + C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (2.44). A consequence of this implication on Eq. (2.55) is that, at a given V, the map parameters should satisfy

$$V = \frac{B + 2C}{B^2 - 4AC} \tag{2.56}$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, substitute Eq. (2.56) in Eq. (2.54),

$$\begin{aligned} (B^2 - 4AC)\left(\frac{B + 2C}{B^2 - 4AC}\right)^2 - 2\frac{B + 2C}{B^2 - 4AC}(A + C) - 1 &= 0 \\ (B + 2C)^2 - 2(B + 2C)(A + C) - (B^2 - 4AC) &= 0 \\ B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC &= 0 \end{aligned}$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (2.56) to

$$A = C \quad \Longrightarrow \quad V = \frac{B + 2C}{B^2 - 4C^2} = \frac{1}{B - 2C}$$

Inserting the latter in Eq. (2.53),

$$\begin{aligned} \left(1 + \frac{2C}{B - 2C}\right) D_v + \frac{B}{B - 2C} S_v &= 0 \\ \frac{B}{B - 2C} D_v + \frac{B}{B - 2C} S_v &= 0 \end{aligned}$$

which implies

$$D_v + S_v = 0 \quad \Longrightarrow \quad v^{(s^*)} = 0$$

Recalling from Tab. 2.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F\left[(c_x - c_y)^2\right]$$

and using the positivity rule of Eq. (2.44) we get

$$V > 0 \quad \text{and} \quad F\left[(c_x - c_y)^2\right] \geq 0 \quad \Longrightarrow \quad v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and viceversa. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \quad \Longrightarrow \quad v^{(s^*)} = v^{(d)} = 0 \quad \Longrightarrow \quad S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v: no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyze similarly the behaviour of the self-consistency equations for $u^{(s^*)}$, $u^{(d)}$. This is done later in Sec. 2.7.4. For now, it suffices to know that limitedly to the sAF phase no gap complexification aforementioned in Sec. 2.4 actually occurs. This effect is not a consequence of MFT derivation, which remains self-coherent and correct; it is instead of the particular form of the non-linear map defined by the set of model equations.

2.7.3 HFPs vanishing in aAF phase

We now deal with a much easier proof for an analogous thesis on the sAF phase described by the five HFPs on the second row of Tab. 2.5, as well as the self-consistency equations of Tabs. 2.3 and 2.4. We continue using the functional F as defined in Eq. (2.42). We also define A, B, C as in Eq. (2.45). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F\left[s_x^2\right] \qquad Q \equiv 2F\left[s_x s_y\right] \qquad R \equiv F\left[s_y^2\right] \tag{2.57}$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A + B + C) - Vu^{(d)}(A - C) \tag{2.58}$$
$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A - C) - Vu^{(d)}(A - B + C) \tag{2.59}$$
$$v^{(p_x)} = -Vv^{(p_x)}P - Vv^{(p_y)}Q \tag{2.60}$$
$$v^{(p_y)} = -Vv^{(p_x)}Q - Vv^{(p_y)}R \tag{2.61}$$

To prove that both $v^{(p_\ell)} = 0$ is immediate.

1. For the aAF phase, $Q = 0$. Let us write Q explicitly

$$\begin{aligned} Q &= F\left[s_x s_y\right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{k_x = -\pi}^{+\pi} \sin k_x \sum_{k_y = -\pi}^{+\pi} \frac{\tilde{\mathrm{Th}}_{(k_x, k_y)}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{(k_x, k_y)}} \sin k_y \end{aligned}$$

From second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function $(\tilde{\mathrm{Th}}/\tilde{E})$ with $\sin k_y$ integrates to zero. This ends the point.

2. $v^{(p_x)} = 0$. To see this, just plug $Q = 0$ in Eq. (2.60),

$$v^{(p_x)} = -Vv^{(p_x)}F\left[s_x^2\right]$$

Using the positivity rule of Eq. (2.44) we get

$$V > 0 \quad \text{and} \quad F\left[s_x^2\right] \geq 0 \quad \Longrightarrow \quad v^{(p_x)} = 0$$

3. $v^{(p_x)} = 0$. Same proof as point above.

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. This is trivially coherent with the constraint of Eq. (2.40). As a consequence, there is no need to distinguish two AF phases,

$$\text{sAF} = \text{aAF}$$

and gap complexification is a completely “ghost” effect, even if theoretically allowed. All is left is to check for the behaviour of $u^{(d)}$.

2.7.4 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \qquad \Longrightarrow \qquad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}$$

Bogoliubov bands maintain renormalization through the $u^{(\gamma)}$ HFPs channel. Aim of this section is to prove that, as was for the normal phase, no nematicity occurs in the AF phase, giving $u^{(d)} = 0$.

While the proof given in Sec. 1.5.2 was long and quite elaborated, this one is actually very short. Let us start from recalling the normal phase context:

$$\text{Normal phase:} \qquad \begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

while, using $\boldsymbol{\pi}$ periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\text{AF phase:} \qquad \begin{aligned} u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \to f'$ exchange.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. 1.5.2 (when showing the same for the normal phase) we never used the specific form of f; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. 1.5.2. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \to x \qquad mU \to \Delta$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)$$

with

$$\begin{aligned} g(x) &= -\frac{x}{\sqrt{x^2 + \Delta^2}} \\ h(x) &= \left(\frac{1}{e^{\beta(\sqrt{x^2+\Delta^2}-\mu)} + 1} - \frac{1}{e^{\beta(-\sqrt{x^2+\Delta^2}-\mu)} + 1} \right) \end{aligned}$$

The following points are true:

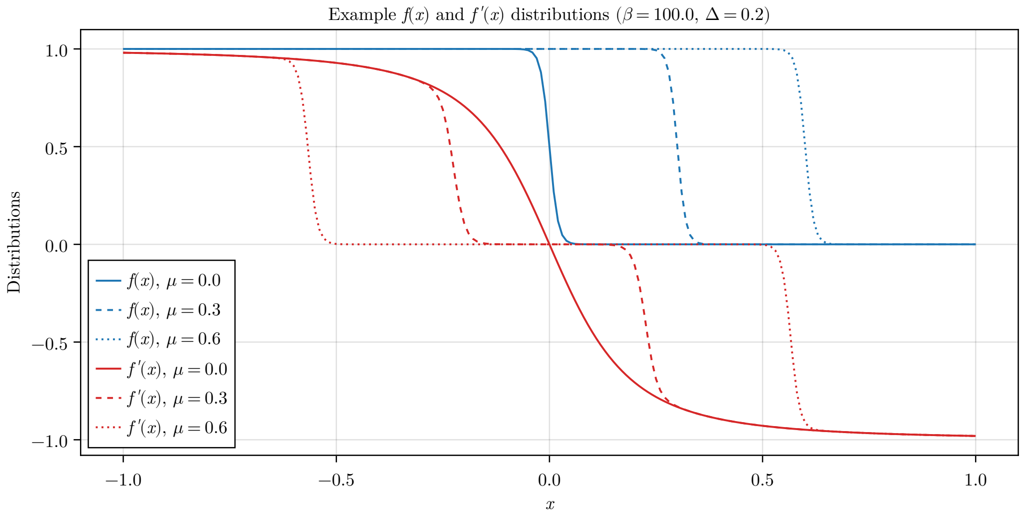


Figure 2.3 Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

1. The function is antisymmetric about $x = 0$. This is evident by considering that $g(x)$ is evidently antisymmetric, while $h(x)$ is symmetric.

2. $g(x)$ is monotonically descendent. Its derivative is

$$\begin{aligned}\frac{\mathrm{d}g}{\mathrm{d}x} &= -\frac{\sqrt{x^2+\Delta^2} - x^2/\sqrt{x^2+\Delta^2}}{x^2+\Delta^2} \\ &= -\frac{\Delta^2}{(x^2+\Delta^2)^{3/2}}\end{aligned}$$

clearly always negative.

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To assert this, we can compute the derivative of f',

$$\frac{\mathrm{d}f'}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}h(x) + g(x)\frac{\mathrm{d}h}{\mathrm{d}x}$$

Now, evidently $h(x) > 0$ and as we said $\mathrm{d}g/\mathrm{d}x < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2+\Delta^2} \qquad \Longrightarrow \qquad \frac{\mathrm{d}E}{\mathrm{d}x} = \frac{x}{E(x)} = -g(x)$$

Then:

$$\begin{aligned}\frac{\mathrm{d}h}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{e^{\beta(-E(x)-\mu)}+1} - \frac{1}{e^{\beta(E(x)-\mu)}+1}\right] \\ &= \beta\frac{\mathrm{d}E}{\mathrm{d}x}\underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{\left[e^{\beta(-E(x)-\mu)}+1\right]^2} + \frac{e^{\beta(E(x)-\mu)}}{\left[e^{\beta(E(x)-\mu)}+1\right]^2}\right]}_{k(x)}\end{aligned}$$

The term in square brackets is obviously positive, $k(x) > 0$. Taking all together, we have

$$g(x)\frac{\mathrm{d}h}{\mathrm{d}x} = -\beta g^2(x)k(x) < 0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}f'}{\mathrm{d}x} < 0$$

making f' descendent monotonicity proven.

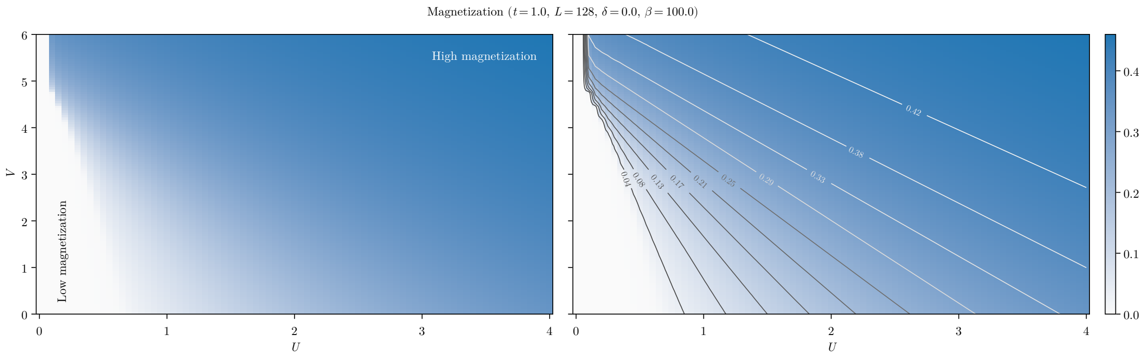


Figure 2.4 Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetization is constant. For the sake of graphic clarity, only the points at $0 \le U \le 4$ and $0 \le V \le 6$ are plotted. [Are $U = 0.05$ data correct?]

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. 1.5.2 without hesitation, and conclude that also for the AF phase

$$u^{(d)} = 0$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (2.41). Starting from a rather complicated scenario, we were able to resolve all the AF HFPs down to the pair

$$m, u^{(s^*)}$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands. Now it is a matter of numeric analysis to assess if Mott localization takes place.

2.8 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \qquad \mathbf{v} = \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

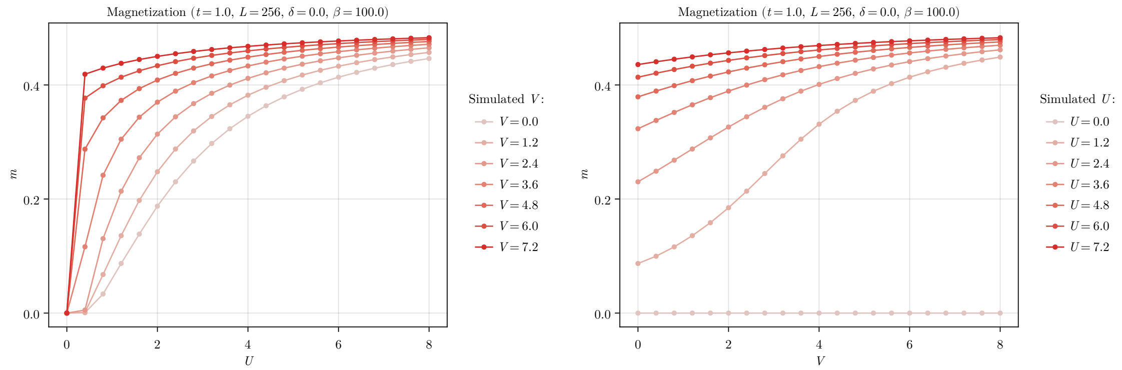
In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0,\ 0.05,\ 0.1,,\ \dots,\ 8.0 \qquad V = 0,\ 0.05,\ 0.1,,\ \dots,\ 8.0 \qquad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 2.5, any finite doping implies immediate disruption of commensurate antiferromagnetism, a consideration that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

2.8.1 Magnetization boost

The convergence values for m computed in the UV plane are shown in the panels of Fig. 2.4, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic



a | U-dependency of m varying V.   b | V-dependency of m varying U.

Figure 2.5 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \qquad \Delta \mathbf{v} = \begin{bmatrix} m \colon 5 \times 10^{-4} \\ u^{(s^*)} \colon 5 \times 10^{-4} \end{bmatrix}$$

The most interesting evident visible directly from heatmaps is that $\hat{H}_V$ actually boosts magnetization. This phenomenon is intuitive and easy to explain: since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

$$\text{EHM: } (t, U, V) \quad \rightarrow \quad \text{HM: } (\tilde{t}, U, 0)$$

being $\tilde{t}$ dependent on V. For the pure HM, to magnetize the system what matters is just U/t. To lower t is physically the same as to increase U, and from this point comes the magnetization precession visible in Fig. 2.4. Isomagnetic lines on the right panel show, for U significantly greater than 0, interesting linear trends whose angular coefficient depend on m itself. The low-U limit of isomagnetic lines presents some oscillations that are evidently imputable to numeric discretization of parametric space. [Add: derivation and discussion about the presence of isomagnetic lines.]

The data shown in Fig. 2.5 confirm clearly the aforementioned magnetization "boost" given by $\hat{H}_V$. The effect is more enhanced at low values of U. Interestingly enough, as is visible from Fig. 2.5b, magnetization does not always saturate the same way. For example, considering the curve at $U = 1.2$, magnetization varies following a somewhat sigmoid shape, making the boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher Us.

2.8.2 Bands shrinking

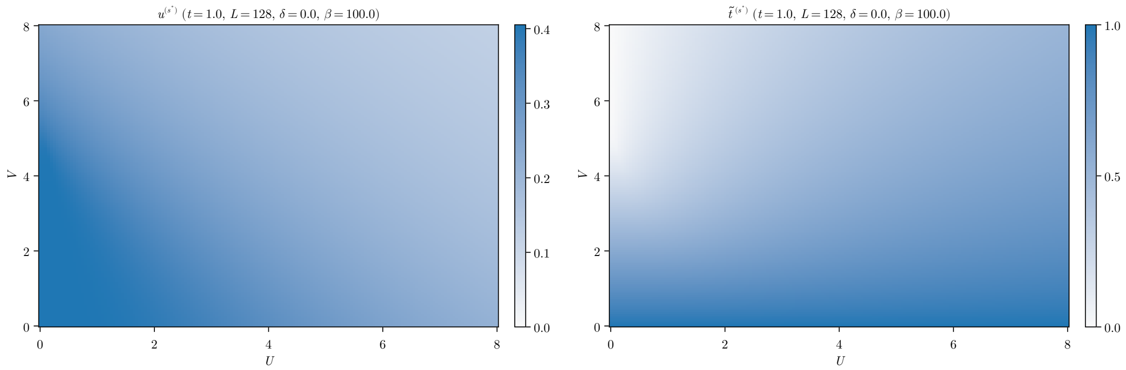
Bands shrinking effect was discussed in Sec. 2.7.1 within C_4 symmetry, and then confirmed in Sec. 2.7.4 when trying to extend to nematic structures. As was for the normal phase, we have proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \quad \rightarrow \quad \tilde{t} < t$$

For the normal phase, in Sec. 1.6.1 we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \rightarrow 0$ limit was approximately $V = 4.93t$, as shown in Fig. 1.7b.

The convergence values for the HFP $u^{(s^*)}$ are shown in Fig. 2.6a, and more in details in the two panels of Fig. 2.7. In general, $u^{(s^*)}$ decreases monotonically both in U and in V. As we already discussed in normal phase, the $U = 0$ data of Fig. 2.7b present a plateau up to a Mott-critical value. This effect is absent at positive Us. Both interactions disrupt the HFP; however, the two limits

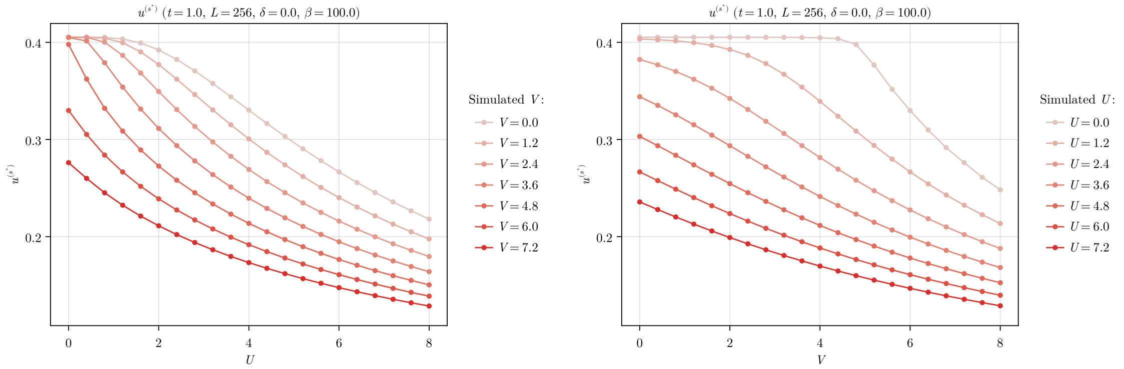
$$U \gg 1 \quad \text{and} \quad V \gg 1$$



a | Heatmap of $u^{(s^*)}$ in UV plane.

b | Heatmap of $t^{(s^*)}$ in UV plane.

Figure 2.6 | Data for the HFP $u^{(s^*)}$ and $\tilde{t}^{(s^*)}$ varying both U and V.



a | U-dependency of $u^{(s^*)}$ varying V.

b | V-dependency of $u^{(s^*)}$ varying U.

Figure 2.7 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

are much different. By computing the value of $\tilde{t}^{(s^*)}$,

$$\tilde{t}^{(s^*)} = 2t - u^{(s^*)}V$$

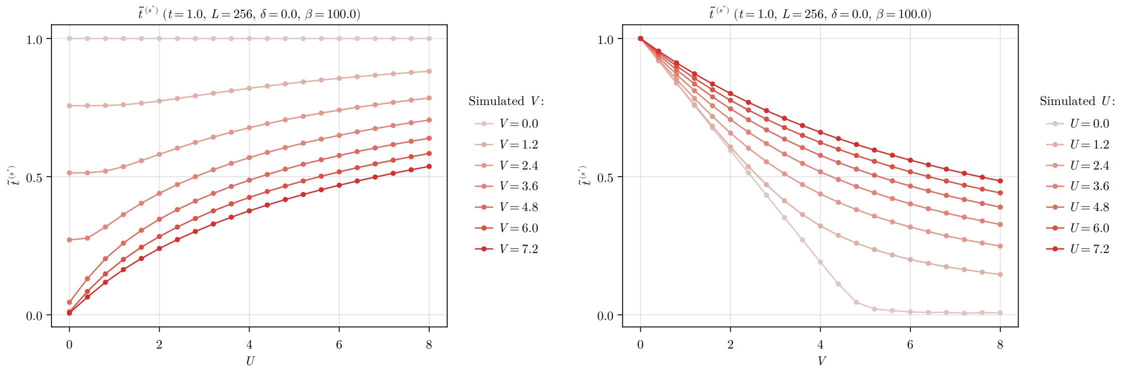
we obtain the data of Fig. 2.6b. The limit $U \gg 1$ taken at fixed V, also shown in Fig. 2.8a, provides "U-induced antiferromagnetism", with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U, also shown in Fig. 2.8b, shrinks bands strongly providing a "Mott-induced antiferromagnetism".

Interestingly, in (U, V) space there is not Mott-localized region of the kind we had in (δ, V) space for the normal phase (see Fig. 1.7b). At half filling, complete localization only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 2.8a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism – we can safely sustain, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

2.8.3 Solution free energy

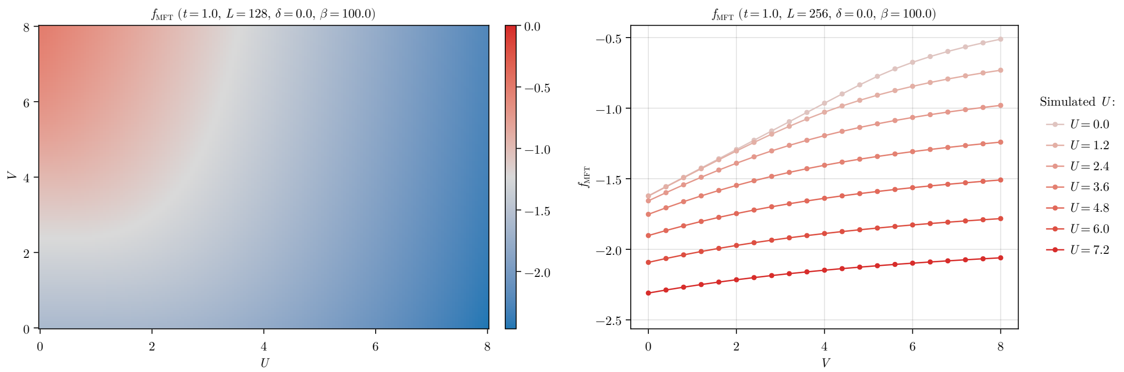
In Fig. 2.9 the solution free energy is plotted. The right panel, Fig. 2.9b, makes it becomes evident how the energetically most stable state is realized at low V, high U. The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 2.9a) instead increases f_{MFT} a lot, correctly reaching zero energy at $U = 0$, $V > 4.93t$ – the critical value discussed in Sec. 1.6.1 and computed in Eq. (1.28).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 2.4, right panel, and thus the realized state exhibits an



a | U-dependency of $t^{(s^*)}$ varying V. b | V-dependency of $t^{(s^*)}$ varying U.

Figure 2.8 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.



a | Heatmap of f_{MFT} in δV plane. b | V-dependency of f_{MFT} varying δ.

Figure 2.9 | Free energy density of the AF phase.

identical value of m, much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localization effect.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. Considering the energy difference,

$$\Delta f_{\mathrm{MFT}} \equiv f_{\mathrm{MFT}}^{(\mathrm{N})} - f_{\mathrm{MFT}}^{(\mathrm{AF})}$$

shown for a slightly more coarse-grained dataset in Fig. 2.10, we see immediately how $\Delta f_{\mathrm{MFT}} \geq 0$ in all (U, V) plane. Note that, in order to compare the two acquisitions, being the normal phase data of Sec. 1.6 having been simulated at $U = 0$ due to their intrinsic U independence, we obtained a comparable dataset for the normal phase by just concatenating the data for $U = 0$ at all values of U we are interested in. As we expect, in the $V \gg 1$ and low U limit the energy difference is null: in this limit the two phases tend to coincide, since at $U = 0.0$ no magnetization is physically possible. Moreover, the $U \gg 1$ limit makes the AF phase undoubtedly energetically favoured. Overall, at half filling the interesting free fermions Mott-localization effect is, at least when comparing these phases, existent but surpassed by the high energetic gain of establishing an AF structure at infinitesimal U. The situation could change at finite doping, a regime we do not investigate limitedly to AF Physics.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic.

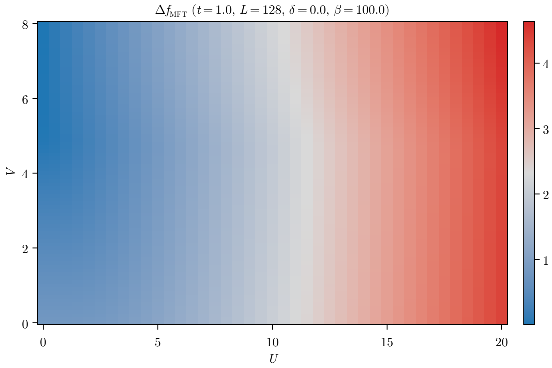


Figure 2.10 | Difference of free energies for the normal phase minus the AF phase.

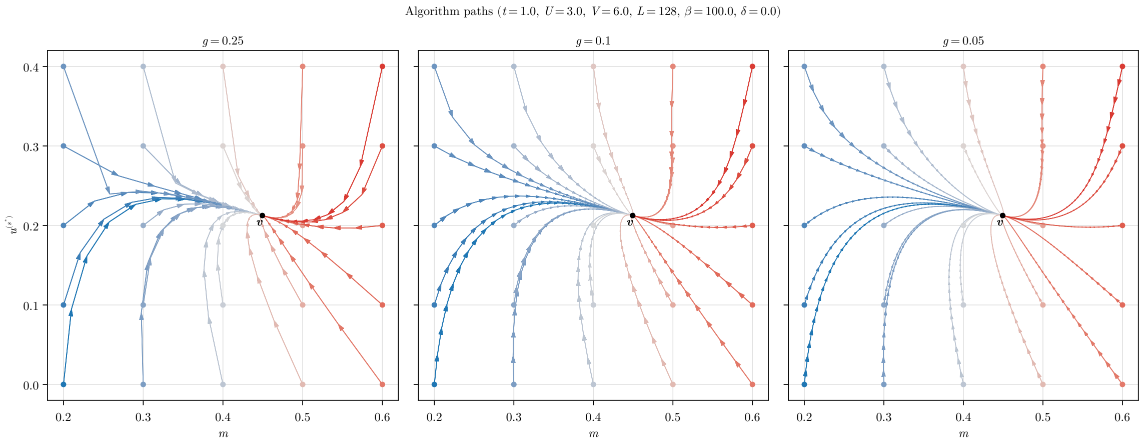


Figure 2.11 | Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

2.8.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. **??** how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}$$

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. 2.7.2, 2.7.3.

In Fig. 2.11 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \qquad V = 6 \qquad L = 128 \qquad \beta = 100 \qquad \delta = 0$$

At this point, the self-consistent HF "fixed-point" solution is given by

$$m \approx 0.449 \qquad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} \quad : \quad m_0 \in \{0.2,\ 0.3,\ ...,\ 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0,\ 0.1,\ ...,\ 0.4\}$$

In figures, different colors indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one step to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g. To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrization could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map "softer" and curvier, as is in the right panel of Fig. 2.11. It's not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Appendix A

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (**??**),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{A.1}$$

are described. The discussion is limited to the two-dimensional square lattice. The results of ths appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

A.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. **??**, AF long-range ordering we deal with here is described by the composite SSB

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^{z}(1)$$
$$\mathrm{T}^{(x)}_{1} \otimes \mathrm{T}^{(y)}_{1} \overset{\mathrm{SSB}}{\to} \mathrm{T}^{(x)}_{2} \otimes \mathrm{T}^{(y)}_{2}$$

Recalling Tab. **??**, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (2.1), (2.2), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.2}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.3}$$

whose EVs are relevant order parameters for the normal and AF phases.

A.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization m, the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{A.4}$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\mathrm{AF})}_{\mathrm{H}/U} \end{aligned} \tag{A.5}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy,

$$\begin{aligned}
E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U\sum_{\mathbf{r}} \langle \hat{n}_{\mathbf{r}\uparrow}\rangle \langle \hat{n}_{\mathbf{r}\downarrow}\rangle \\
&= U\sum_{\mathbf{r}} \left(n+(-1)^{x+y}m\right)\left(n-(-1)^{x+y}m\right) \\
&= L_xL_y \times U\left(n^2-m^2\right)
\end{aligned} \tag{A.6}$$

By a Fourier transform, the phase factor of Eq. (A.5) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned}
\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}}(-1)^{x+y}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}
\end{aligned} \tag{A.7}$$

where $\boldsymbol{\pi}=(\pi,\pi)$. It follows:

$$mU\sum_{\mathbf{r}}(-1)^{x+y}\left[\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right] = mU\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow}-\hat{c}^\dagger_{\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU\sum_{\mathbf{r}}\left[\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right] = nU\times\hat{N} \qquad \Longrightarrow \qquad \mu\to\tilde{\mu}\equiv\mu-nU$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (??),

$$-t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma}+\mathrm{h.c.}\right] \overset{\mathcal{F}}{=} \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (??).

Take everything together. Using the property:

$$\begin{aligned}
\sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\right]
\end{aligned} \tag{A.8}$$

and reduce Eq. (A.5) to its reciprocal form

$$\hat{H}\simeq\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\left[\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]-mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right]+nU\hat{N}-E^{(\mathrm{AF})}_{\mathrm{H}/U}$$

A.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$\begin{aligned}
\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}-\mathrm{d.c.} &= \hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} \\
&= \hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow} \\
&= \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right)-\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right)
\end{aligned}$$

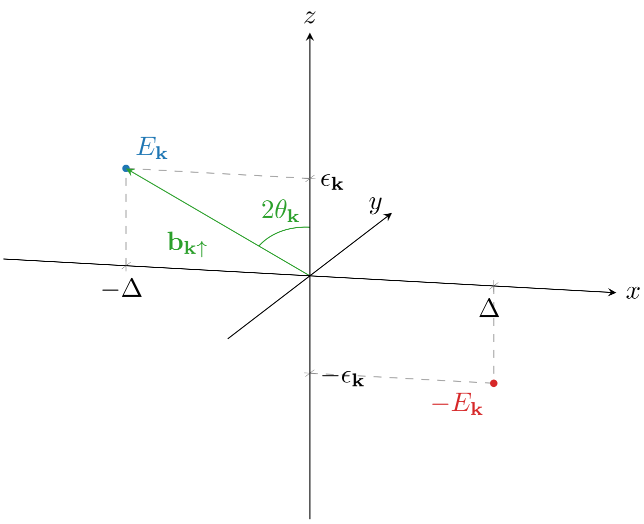


Figure A.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma =\uparrow$ situation is plotted.

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle + \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y}\frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle - \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are $\mathbf{r}$-independent indeed, it must hold

$$n = \frac{1}{L_x L_y}\sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle + \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \tag{A.9}$$

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{r}} (-1)^{x+y}\frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle - \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \tag{A.10}$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

A.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \to \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^{*}_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.11}$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma}\tau^z - \Delta_{\mathbf{k}\sigma}\tau^x$$

being τ^α the Pauli matrices. Then, defining

$$\hat{s}^\alpha_{\mathbf{k}\sigma} \equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^\alpha\hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_\sigma \mathbf{b}_{\mathbf{k}\sigma}\cdot\hat{\mathbf{s}}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.12}$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. A.1. Evidently the following relations, also reported in Tab. 2.2, hold:

$$\hat{s}^x_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \tag{A.13}$$
$$\hat{s}^y_{\mathbf{k}\sigma} = i\left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) \tag{A.14}$$
$$\hat{s}^z_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{A.15}$$

Let also:

$$\begin{aligned}\hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\mathbb{I}_{2\times2}\hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{aligned} \tag{A.16}$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}h_{\mathbf{k}\sigma}W^\dagger_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned}W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix}\end{aligned} \tag{A.17}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \tag{A.18}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of “down” spins with “up” spins. In the pseudospin picture, at each “site” $(\mathbf{k}\sigma)$, the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{A.19}$$

that will appear quite often. Then it’s immediate to derive the various expectation values.

$$\langle\hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.20}$$
$$\langle\hat{s}^y_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.21}$$
$$\langle\hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.22}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (A.16) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \quad\Longrightarrow\quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times2}\right]W_{\mathbf{k}\sigma}^{\dagger} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times2} \tag{A.23}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \tag{A.24}$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned}\hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \\ \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix}\end{aligned} \tag{A.25}$$

The consequence of Eq. (A.23) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. **??** retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, imposing that the chemical potential crosses the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where the chemical potential lies within the gap. As will be detailed in A.1.8, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [15].

A.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \to +\infty$. Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudospin states. Let $\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\text{MBZ}}\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle \tag{A.26}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}}\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^+_{\mathbf{k}\sigma} - \hat{s}^-_{\mathbf{k}\sigma})\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}\left|\Downarrow_{\mathbf{k}\sigma}\right\rangle - \sin\theta_{\mathbf{k}\sigma}\left|\Uparrow_{\mathbf{k}\sigma}\right\rangle\end{aligned} \tag{A.27}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^x_{\mathbf{k}\sigma} \pm i\hat{s}^y_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (A.16) and (A.15),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get immediately

$$\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle=1 \qquad \frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle=0$$
$$\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle=0 \qquad \frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle=1$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle\sim\hat{c}^\dagger_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle\sim\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.26) and (A.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}+e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.28}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle&=\left(e^{2i\zeta_{\mathbf{k}\sigma}}+e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma}\\&=\cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

but from Eq. (A.13)

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle&=\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle\\&=\sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad\Longrightarrow\quad \zeta_{\mathbf{k}\sigma}=k\pi \quad \text{with} \quad k\in\mathbb{Z}$$

and lets us conclude from Eq. (A.28)

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}+\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.29}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system. Finally, using Eq. (A.25), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.30}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

A.1.5 Spin degeneracy removal

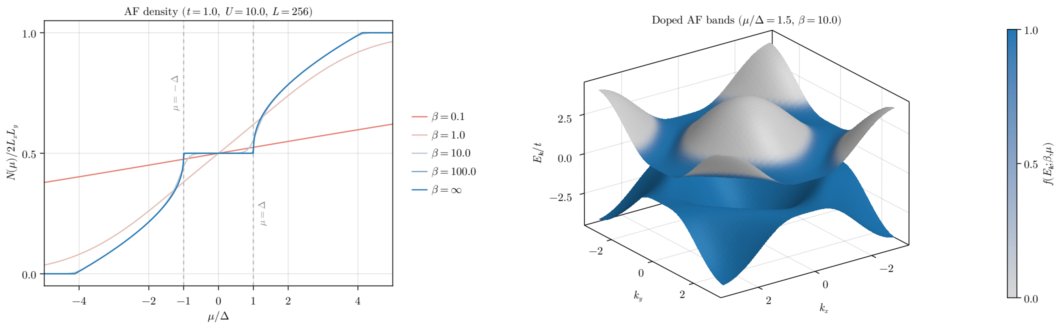
Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow}=\epsilon_{\mathbf{k}\downarrow} \qquad \Delta_{\mathbf{k}\uparrow}=-\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow}=E_{\mathbf{k}\downarrow}$. Since the gap function sign does has not a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma=\uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow}=mU\equiv\Delta$$

being the latter purely s-wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.



a | Electronic density.   b | Example doped bands.

Figure A.2 | Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

A.1.6 Chemical potential and system density

Using the fact that

$$\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle = \left\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \right\rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (A.9))

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle \\ &= 2 \sum_{\mathbf{k}\in\mathrm{MBZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \end{aligned} \tag{A.31}$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ,

$$f(\epsilon; \beta, \mu) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\boldsymbol{\pi}}$$

it holds equivalently

$$N = \sum_{\mathbf{k}\in\mathrm{BZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \tag{A.32}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, MBZ $\to$ BZ. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m, which is just the AF instability order parameter.

In Fig. A.2 the density $n(\mu)$ is plotted as a function of μ. As is evident, at nearly zero temperature ($\beta \to \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

A.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (A.7), (A.8) and the definition of Eq. (A.3), we get

$$\frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]$$

thus employing Eq. (A.10)

$$\begin{aligned}m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\rangle\\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^{\dagger}_{\mathbf{k}\downarrow}\right)\right\rangle\end{aligned}$$

Using now Eq. (A.13) we get the self-consistency equation for the HFP m,

$$\begin{aligned}m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right]\\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}-\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu]\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu]\end{aligned}\tag{A.33}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s-wave component.

Consider specifically the half-filled situation and multiply Eq. (A.33) on both sides by U. Since due to symmetry $\mu=0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

where we used the relation

$$\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L\to\infty$

$$\frac{1}{U}\simeq\int_{\mathrm{MBZ}}\frac{\mathrm{d}\mathbf{k}}{(2\pi)^2}\frac{1}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

Substitute the momentum integral with an integral over energies, in the low temperature limit, $\beta\to\infty$ (thus approximate the hyperbolic tangent with 1)

$$\frac{1}{U}\simeq\int_{\mathrm{bands}}\frac{\mathrm{d}\epsilon\rho(\epsilon)}{\sqrt{\epsilon^2+\Delta^2}}$$

[To be continued...]

A.1.8 Instability of the commensurate AF solution and atypical metallic bands

A key point to be discussed is the known instability of the AF HF solution of the EHM. As is thoroughly discussed in [15], the solution we presented above is only stable at half-filling, becoming hardly unstable at infinitesimal doping. This is due to the commensurate nature of the established AF phase: we impose SSB to take place at wavevector $\boldsymbol{\pi}$, shrinking the BZ to exactly half its size and giving back on the newly defined MBZ two bands $\pm E_{\mathbf{k}}$. This procedure actually works well for the half-filled case: the magnetization essentially depends on the scattering rate between points on the FS connected by a $\boldsymbol{\pi}$ vector. The half filled case, taking advantage of the natural nesting property of the bare bands,

$$\forall \mathbf{k} \in \partial\text{MBZ} : \; \epsilon_{\mathbf{k}} = \epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

(being ∂MBZ the boundary of the MBZ) exhibits a magnetization at infinitesimal interaction. For a many body state resulting from continuous SSB, to establish a given phase means to show the presence of a stable gapless Goldstone mode associated to the broken symmetry. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the AF phase, it follows from the reduced $\mathrm{U}^z(1)$ rotational symmetry and is related to the proper transverse pseudospin-pseudospin response function, 2×2 matrix equation,

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - U\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{A.34}$$

being χ the free transverse response. The pseudospin hereby considered is the one of Eq. (A.12). As is shown in [15], for the commensurate AF SSB at wavevector $\boldsymbol{\pi}$ the condition for the presence of a stable gapless mode reduces to a 2×2 matrix equation,

$$\text{Stability condition:} \qquad \mathbb{I}_{2\times 2} - U\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},0)\right] \geq 0 \tag{A.35}$$

with $\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},\omega)\right]$ the response function matrix obtained by considering a matrix response function for a two-sites unit cell. The above equation is to be interpreted as a matricial equation [Explain better..]. Considering the eigenvalue of largest amplitude of the matrix λ_δ at filling δ, the one most prone to instabilities, it is found that

$$\begin{aligned} \lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} && \text{(half filling)} \\ \lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} + \frac{2\epsilon_{\mathrm{F}}}{\Delta}N(\epsilon_{\mathrm{F}}) && \text{(finite doping)} \end{aligned}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. At half-filling the system is an insulator, therefore no Fermi energy is defined, as well as no FS. Evidently this implies

$$\begin{aligned} 1 - U\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) &= 0 \quad \text{(half filling)} && \Longrightarrow \quad \text{stable Goldstone mode} \\ 1 - U\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) &< 0 \quad \text{(finite doping)} && \Longrightarrow \quad \text{unstable Goldstone mode} \end{aligned}$$

For any finite doping, the commensurate AF phase becomes unstable and is destroyed by transverse spin fluctuations. Nevertheless, for the sake of completeness and because we noticed too late, we will study the metallic AF also at finite doping.

A.1.9 Hartree-Fock algorithm results in reciprocal space

The above sections offer a self-consistency equation for the only HFP m; W is determined by $\Delta = mU$ indeed. In an iterative HF fashion, Eq. (A.33) reads:

$$m_{i+1} = \frac{U}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]} \left[f(-E_{\mathbf{k}}[m_i];\beta,\mu) - f(E_{\mathbf{k}}[m_i];\beta,\mu) \right] \tag{A.36}$$

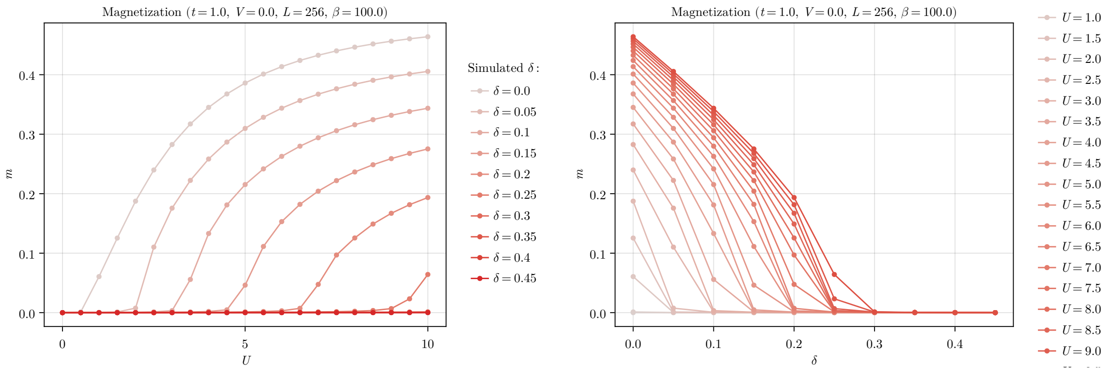
Then a self-consistent algorithm to search for a self consistent estimate of the magnetization may be sketched, following the general idea of Sec. ??:

1. Algorithm setup: initialize a counter $i = 1$ and choose:
   - the local repulsion U/t or equivalently the hopping amplitude t/U;
   - the coarse-graining of the BZ, which is, fix L_x and L_y. Then
   $$k_x = 2\pi n_x/L_x \qquad k_y = 2\pi n_y/L_y$$
   where $-L_j/2 \leq n_j \leq L_j/2$, $n_j \in \mathbb{Z}$;
   - the density n or equivalently the doping $\delta \equiv n - 0.5$;
   - the inverse temperature β;
   - the maximum number of iterations $p \in \mathbb{N}$;
   - the mixing parameter $g \in [0, 1]$;
   - the tolerance parameters $\delta_m, \delta_n \in \mathbb{R}$ (respectively for the HFP m and density);
2. Select a random starting value $m_i > 0$ for $i = 1$;
3. Find the optimal chemical potential μ as follows:
   a) Compute $\pm E_{\mathbf{k}}[\Delta_i, U]$;
   b) Define the function of the chemical potential μ,
   $$\delta n(\mu; \beta, m_i, U) \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[f\left(-E_{\mathbf{k}}[U, m_i]; \beta, \mu\right) + f\left(E_{\mathbf{k}}[U, m_i]; \beta, \mu\right)\right] - n$$
   c) Find the root of δn,
   $$\delta n(\mu_i) = 0$$
   Then μ_i is the chemical potential which, at step i with HFP m_i, realizes the best approximation of density n;
4. Compute m_{i+1} using Eq. (A.36) with chemical potential μ_i and update the counter, $i \to i+1$;
5. Check if $|m_{i+1} - m_i| \leq \delta_m$:
   - If yes, halt;
   - If not: check if $i > p$
     - If yes, halt and consider choosing better tolerance and model parameters;
     - If not, define
     $$m_{i+1} \colon g m_{i+1} + (1 - g) m_i$$
     (logical assignment notation used) and repeat from step 2.

The algorithm sketched hereby was ran with the following setup:

```
UU = [U for U in 0.5:0.5:20.0]        # Local repulsions
LL = [256]                             # Lattice sizes
dd = [d for d in 0.0:0.05:0.45]        # Dopings
bb = [100.0, Inf]                      # Inverse temperatures
p = 100                                # Max number of iterations
dm = 1e-4                              # Tolerance on magnetization
dn = 1e-2                              # Tolerance on density
g = 0.5                                # Mixing parameter
```

In Fig. A.3 plots at $\beta = 100$ of the magnetization m is reported. As is evident in Fig. A.3a, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger, a mere consequence of the fact that the free space at each single-particle state in k-space shrinks as density increases; also, for $\delta > 0$, a finite magnetization m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping). Fig. A.3b shows the dependence of magnetization on doping at various fixed local repulsions U: once again, to dope the material disrupt AF ordering. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures (predictably).



a | Local repulsion U/t depencence. b | Doping $\delta = n - 0.5$ dependence.

Figure A.3 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. A.3a) and the doping $\delta = n - 1/2$ (Fig. A.3b).

A.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. A.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{i\bar{\sigma}} \rangle$$

which, having we broken $U^{\mathrm{c}}(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. **??** that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

A.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. A.1.1 simply by setting $m = 0$. We immediately get from Eq. (A.5):

$$\begin{aligned} \hat{H}_U &\simeq U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] + (\text{Cooper channel}) \\ &= nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \end{aligned} \tag{A.37}$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (A.6),

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{A.38}$$

We now move to Cooper channel.

A.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (**??**) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big] \tag{A.39}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (**??**) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{A.40}$$

and define $w_{\mathbf{q}}$ as in Eq. (**??**),

$$w_{\mathbf{q}} \equiv \langle \phi^\dagger_{\mathbf{q}} \rangle$$

Its s-wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} w_{\mathbf{q}} \tag{A.41}$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (A.2). Eq. (A.37) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \Big[\langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}^\dagger_{\mathbf{q}} \langle \hat{\phi}_{\mathbf{q}'} \rangle \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.42}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy,

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}\mathbf{q}'} \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U \big| w^{(s)} \big|^2 \end{aligned} \tag{A.43}$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q}\in\mathrm{BZ}} \Big[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.44}$$

which can be easily mapped on a pseudospin model.

Composing Eqns. (A.37) and (A.44), we get the full MFT hamiltonian

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \Big[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \Big] - \sum_{\mathbf{q}\in\mathrm{BZ}} \Big[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] + nU \times \hat{N} - \Big[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \Big] \tag{A.45}$$

A.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. A.1.3. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

Define the pseudospin operator:

$$\hat{s}^\alpha_{\mathbf{k}} \equiv \hat{\Phi}^\dagger_{\mathbf{k}} \tau^\alpha \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^*-wave structure, since no Fock interaction in the present context chan change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}^\dagger_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}^\dagger_{\mathbf{k}}\tau^z\hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \tag{A.46}$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \left(\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}\right)\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right] \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\hat{s}^z_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} \end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the "anticommutation energy" coming from the kinetic term,

$$E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{A.47}$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (A.42),

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}}\hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{A.48}$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \qquad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}} \qquad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{\mathrm{Re}\{\Delta_{\mathbf{k}}\}} \tag{A.49}$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}}\cdot\hat{\mathbf{s}}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}} \\ \hat{s}^y_{\mathbf{k}} \\ \hat{s}^z_{\mathbf{k}} \end{bmatrix} \tag{A.50}$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (A.17). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_{\mathbf{k}}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_{\mathbf{k}}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{A.51}$$

obtaining the diagonal matrix

$$d_{\mathbf{k}} \equiv W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^\dagger \quad \text{where} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}$$

and the bands

$$E_{\mathbf{k}} \equiv \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (A.24)

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} \tag{A.52}$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix} \end{aligned} \tag{A.53}$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_{\mathbf{k}}$.

A.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. A.1.4, we can extract the half-filling zero temperature ground state, commonly referred to as the “BCS state”. Eqns. (A.26) and (A.27) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} |\Downarrow_{\mathbf{k}}\rangle - \sin\theta_{\mathbf{k}} |\Uparrow_{\mathbf{k}}\rangle\right]$$

Using now Eq. (A.46), we have

$$\begin{aligned} \langle \Downarrow_{\mathbf{k}} | \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} | \Downarrow_{\mathbf{k}} \rangle &= 0 \\ \langle \Uparrow_{\mathbf{k}} | \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} | \Uparrow_{\mathbf{k}} \rangle &= 2 \end{aligned}$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}^\dagger_{\mathbf{k}} |\Omega_{\mathbf{k}}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}} \sin\theta_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}}\right] |\Omega_{\mathbf{k}}\rangle \tag{A.54}$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

A.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (A.13) and (A.14):

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{A.55}$$

$$\hat{s}^y_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}\right) \tag{A.56}$$

while the analogous of Eq. (A.15) was given in Eq. (A.46). Let also the thermal factor

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{A.57}$$

and the EVs:

$$\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.58}$$

$$\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.59}$$

$$\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.60}$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (A.41): explicitly

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i\,\langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ &= \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{A.61}$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -Uw^{(s)}$, thus if we define

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}}\right]^{-1}$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]}w^{(s)}$$

Since evidently $U_{\mathrm{c}}[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

Bibliography

[1] James F. Annett. “Symmetry of the order parameter for high-temperature superconductivity”. In: *Advances in Physics* 39.2 (Apr. 1990). Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/00018739000101481, pp. 83–126. ISSN: 0001-8732. DOI: 10.1080/00018739000101481. URL: https://doi.org/10.1080/00018739000101481 (visited on 01/06/2026).

[2] Daniel P. Arovas et al. “The Hubbard Model”. In: *Annual Review of Condensed Matter Physics* 13.Volume 13, 2022 (2022), pp. 239–274. ISSN: 1947-5462. DOI: https://doi.org/10.1146/annurev-conmatphys-031620-102024. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-conmatphys-031620-102024.

[3] Anna E. Böhmer et al. “Nematicity and nematic fluctuations in iron-based superconductors”. en. In: *Nature Physics* 18.12 (Dec. 2022), pp. 1412–1419. ISSN: 1745-2481. DOI: 10.1038/s41567-022-01833-3. URL: https://www.nature.com/articles/s41567-022-01833-3.

[4] Zhangkai Cao et al. “Dominant p-wave pairing induced by nearest-neighbor attraction in the square-lattice extended Hubbard model”. In: *Phys. Rev. B* 111 (2 Jan. 2025), p. 024509. DOI: 10.1103/PhysRevB.111.024509. URL: https://link.aps.org/doi/10.1103/PhysRevB.111.024509.

[5] Piers Coleman. *Introduction to Many-Body Physics*. Cambridge University Press, 2015.

[6] Michele Fabrizio. *A Course in Quantum Many-Body Theory*. Springer, 2022.

[7] Gabriele Giuliani and Giovanni Vignale. *Quantum Theory of the Electron Liquid*. Cambridge University Press, 2005.

[8] Giuseppe Grosso and Giuseppe Pastori Parravicini. *Solid State Physics*. Second Edition. Academic Press, 2014.

[9] J. E. Hirsch. “Two-dimensional Hubbard model: Numerical simulation study”. In: *Phys. Rev. B* 31 (7 Apr. 1985), pp. 4403–4419. DOI: 10.1103/PhysRevB.31.4403. URL: https://link.aps.org/doi/10.1103/PhysRevB.31.4403.

[10] Joel Hutchinson and Frank Marsiglio. “Mixed temperature-dependent order parameters in the extended Hubbard model”. en. In: *Journal of Physics: Condensed Matter* 33.6 (Feb. 2021), p. 065603. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/1361-648X/abc801. URL: https://iopscience.iop.org/article/10.1088/1361-648X/abc801 (visited on 01/06/2026).

[11] R. Micnas, J. Ranninger, and S. Robaszkiewicz. “Superconductivity in narrow-band systems with local nonretarded attractive interactions”. In: *Reviews of Modern Physics* 62.1 (Jan. 1990). Publisher: American Physical Society, pp. 113–171. DOI: 10.1103/RevModPhys.62.113. URL: https://link.aps.org/doi/10.1103/RevModPhys.62.113 (visited on 01/06/2026).

[12] Robin Scholle et al. “Comprehensive mean-field analysis of magnetic and charge orders in the two-dimensional Hubbard model”. In: *Phys. Rev. B* 108 (3 July 2023), p. 035139. DOI: 10.1103/PhysRevB.108.035139. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.035139.

[13] P. Senarath Yapa et al. "Mixed-symmetry superconductivity and the energy gap". en. In: *Physica C: Superconductivity and its Applications* 633 (June 2025), p. 1354719. ISSN: 09214534. DOI: 10.1016/j.physc.2025.1354719. URL: https://linkinghub.elsevier.com/retrieve/pii/S0921453425000723 (visited on 01/06/2026).

[14] Manfred Sigrist and Kazuo Ueda. "Phenomenological theory of unconventional superconductivity". In: *Reviews of Modern Physics* 63.2 (Apr. 1991). Publisher: American Physical Society, pp. 239–311. DOI: 10.1103/RevModPhys.63.239. URL: https://link.aps.org/doi/10.1103/RevModPhys.63.239 (visited on 01/07/2026).

[15] Avinash Singh and Zlatko Tešanović. "Collective excitations in a doped antiferromagnet". en. In: *Physical Review B* 41.1 (Jan. 1990), pp. 614–631. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.41.614. URL: https://link.aps.org/doi/10.1103/PhysRevB.41.614 (visited on 12/12/2025).

[16] Ashvin Vishwanath. "Dirac Nodes and Quantized Thermal Hall Effect in the Mixed State of d-wave Superconductors". In: *Physical Review B* 66.6 (Aug. 2002). arXiv:cond-mat/0104213, p. 064504. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.66.064504. URL: http://arxiv.org/abs/cond-mat/0104213 (visited on 01/05/2026).

[17] Alexander Wietek et al. "Stripes, Antiferromagnetism, and the Pseudogap in the Doped Hubbard Model at Finite Temperature". en. In: *Physical Review X* 11.3 (July 2021), p. 031007. ISSN: 2160-3308. DOI: 10.1103/PhysRevX.11.031007. URL: https://link.aps.org/doi/10.1103/PhysRevX.11.031007 (visited on 12/12/2025).

[18] C. Zhou and H. J. Schulz. "Density of states and tunneling spectra in two-dimensional d-wave superconductors". In: *Physical Review B* 45.13 (Apr. 1992). Publisher: American Physical Society, pp. 7397–7401. DOI: 10.1103/PhysRevB.45.7397. URL: https://link.aps.org/doi/10.1103/PhysRevB.45.7397 (visited on 01/06/2026).